Nonparametric Statistics

Lecture notes based on Jung, *Nonparametric Statistics with R* (Ch. 1–12)

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Chapter 1. Introduction to Nonparametric Statistics

1.1 Parametric and nonparametric models

A statistical model is a family $\mathcal{F}$ of distributions assumed to contain the population. It is **parametric** if it is indexed by finitely many real numbers, e.g. the normal family

$$\mathcal{F}_G = \{N(\mu, \sigma^2) : \mu \in \mathbb{R},\ \sigma > 0\}. \tag{1.1}$$

It is **nonparametric** if no finite-dimensional parametrization exists. Typical examples:

$$\mathcal{F}_1 = \{F : \int x^2\, dF < \infty\}, \qquad \mathcal{F}_{\text{all}} = \{\text{all } F\},$$

$$\mathcal{F}_2 = \{F : F \text{ has a density symmetric about some } \theta\}.$$

Both $\mathcal{F}_G \subset \mathcal{F}_1$ and $\mathcal{F}_G \subset \mathcal{F}_2$. A model with a finite-dimensional part of interest and an infinite-dimensional nuisance part, such as $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ with $\epsilon_i \overset{\text{i.i.d.}}{\sim} F$ unrestricted, is called **semi-parametric**.

In a nonparametric model a **parameter** is a functional $\theta = \theta(F)$: the mean $\int x\, dF$, the median, a quantile, the density f, or a regression curve $m(x) = E(Y \mid X = x)$. Knowing a few such values never determines F. A procedure whose validity (e.g. type I error) holds for every F in a very large family is called **distribution-free**.

The course uses three tools:

1. **Ranks, signs and permutation tests** (Part I): conditional inference given the observed values, exact under exchangeability.
2. **The bootstrap** (Part II): replace the unknown F by an estimate and compute sampling distributions by simulation.
3. **Smoothing** (Part II): estimate a function $f(x)$ or $m(x)$ by averaging the observations near x.

1.2 Review of parametric inference

Motivation. Nonparametric methods are best understood as answers to the failures of parametric ones. We recall the t-test and the likelihood framework so that later we can see exactly which assumptions each rank method drops and what it costs in efficiency.

Definition (Quantile)

For a distribution function F and $p \in (0,1)$, the p-quantile is $Q(p) = \min\{x : F(x) \geq p\}$. For the standard normal, $z_\alpha := \Phi^{-1}(1-\alpha)$.

Proposition 1.1

Let F be the distribution function of an integer-valued X (e.g. $X \sim B(n,p)$). Then $q = Q(p)$ is the smallest integer with $F(q) \geq p$, and $F(q-1) < p \leq F(q)$. In particular $P(X \leq q) \geq p$ and $P(X \geq q) > 1 - p$.

Proof. F is nondecreasing, right-continuous and constant on each $[k, k+1)$, so $\{x : F(x) \geq p\} = [q, \infty)$ with q the smallest integer satisfying $F(q) \geq p$. Minimality gives $F(q-1) < p$. Then $P(X \geq q) = 1 - F(q-1) > 1 - p$. ∎

Two reference distributions recur. If $X \sim N(\mu, \sigma^2)$ then $(X-\mu)/\sigma \sim N(0,1)$, so $F_X(x) = \Phi((x-\mu)/\sigma)$. If $X \sim B(n,p)$ (number of successes in n independent trials), then $EX = np$, $\operatorname{Var} X = np(1-p)$ (a sum of n Bernoulli(p) variables), and for large n (roughly $np \geq 10$) X is approximately $N(np, np(1-p))$ by the CLT below.

Estimation. An estimator $\hat{\theta}$ is judged by its bias $E\hat{\theta} - \theta$, standard error $\operatorname{se}(\hat{\theta}) = \sqrt{\operatorname{Var}\hat{\theta}}$ and mean squared error

$$\operatorname{MSE}(\hat{\theta}) = E(\hat{\theta}-\theta)^2 = \operatorname{bias}(\hat{\theta})^2 + \operatorname{Var}(\hat{\theta}).$$

Under $\mathcal{F}_G$, $\bar{X}_n$ is the best unbiased estimator of μ, and $T = \sqrt{n}(\bar{X}_n - \mu)/s \sim t_{n-1}$ gives the exact interval $\bar{X}_n \pm t_{\alpha/2,n-1} s/\sqrt{n}$. Outside $\mathcal{F}_G$ these optimality and exactness statements are lost.

Testing. A test of H_0 against H_1 rejects when a statistic T falls in a rejection region chosen so that $P_{H_0}(\text{reject}) \leq \alpha$ (the **level**); among such tests we want large **power** $P_{H_1}(\text{reject})$. The **p-value** is the probability under H_0 of a value of T at least as extreme (in the direction of H_1) as the observed one; rejecting when $p \leq \alpha$ is the same as using the level-α rejection region. In nonparametric problems the central task is to find a statistic whose **null distribution** does not depend on the unknown F.

Proposition 1.2 (Duality of tests and intervals)

Suppose for each θ_0 there is a level-α test ϕ_{θ_0} of $H_0 : \theta = \theta_0$. Then $C = \{\theta_0 : \phi_{\theta_0} \text{ does not reject}\}$ is a $1-\alpha$ confidence set. Conversely, if C is a $1-\alpha$ confidence set, rejecting $H_0 : \theta = \theta_0$ when $\theta_0 \notin C$ is a level-α test.

Proof. If θ is the true value, $P(\theta \in C) = P(\phi_\theta \text{ does not reject}) \geq 1-\alpha$. Conversely $P_{\theta_0}(\theta_0 \notin C) \leq \alpha$. ■

The central limit theorem. Much of classical practice relies on large-sample normality.

Theorem 1.3 (CLT)

If $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F$ with mean μ and variance $0 < \sigma^2 < \infty$, then $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} N(0,1)$.

Proof [Supplement]. Let $Y_i = (X_i - \mu)/\sigma$ and $\varphi(t) = Ee^{itY_1}$. Since $EY_1^2 < \infty$, $\varphi(t) = 1 - \frac{t^2}{2} + o(t^2)$. By independence $Ee^{itZ_n} = \varphi(t/\sqrt{n})^n = \left(1 - \frac{t^2}{2n} + o(\frac{1}{n})\right)^n \to e^{-t^2/2}$, the characteristic function of $N(0,1)$; Lévy's continuity theorem concludes. ■

Because of the CLT, t-based procedures are routinely applied to discrete, ordinal or skewed data: what matters is that the *statistic*, not the population, is approximately normal. The CLT says nothing about how large n must be. Convergence is slow for skewed F and fails entirely when $\sigma^2 = \infty$ (e.g. Cauchy or slash populations). This is the first motivation for nonparametric methods.

1.3 Why nonparametric methods

- *Validity under weak assumptions.* Rank, sign and permutation tests control the type I error exactly for every F in a large family, at any sample size.
- *Computation instead of derivation.* Permutation and bootstrap distributions replace formulas that are unknown or valid only asymptotically.
- *Flexibility.* Function-valued targets (f, m) are estimated without assuming a shape.

- *Robustness.* Procedures based on ranks, signs or medians are insensitive to outliers.

Example 1.1 (Effect of one outlier)

Take the 8 values 2.1, 2.4, 2.6, 2.9, 3.0, 3.2, 3.5, 25. With the last value, the mean is 5.59 and the standard deviation 7.86; without it, 2.81 and 0.48. The median is 2.95 with it and 2.9 without. The **median absolute deviation** MAD $= \text{median}_i |x_i - \text{median}_j\, x_j|$ is 0.45 with it and 0.30 without. The mean and standard deviation are dominated by a single point; the median and MAD barely move.

An outlier can be viewed as a draw from a heavy-tailed population. Distribution-free methods keep their properties under heavy tails, so they are typically robust (the converse need not hold).

Chapter 2. Distribution-Free Inference for One Population

Setting

$X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} F$ with median θ. We test $H_0 : \theta = \theta_0$ (against $\theta > \theta_0$, $\theta < \theta_0$ or $\theta \neq \theta_0$), estimate θ, and give confidence intervals, under two assumptions of increasing strength: F continuous (sign test), and F continuous and symmetric about θ (signed-rank test).

2.1 The sign test

Motivation. If we know nothing about the distribution except that it is continuous, the number of observations above the hypothesized median is still exactly binomial. The sign test uses only this, so it is valid for any distribution.

Definition (Sign statistic)

$S = S(\theta_0) = \sum_{i=1}^{n} I(X_i > \theta_0)$, the number of observations above θ_0.

Proposition 2.1

If $P(X_i = \theta_0) = 0$ and θ_0 is the median of F, then $S \sim B(n, \frac{1}{2})$ for every such F.

Proof. The indicators $I(X_i > \theta_0)$ are i.i.d. Bernoulli with success probability $P(X_1 > \theta_0) = 1 - F(\theta_0) = \frac{1}{2}$. ■

Thus the sign test is distribution-free. Under $\theta > \theta_0$ each sign is "+" with probability $p_1 = P(X > \theta_0) > \frac{1}{2}$, so we reject for large S. The upper-tailed p-value is $P(S \geq s_{\text{obs}})$; the two-sided one is $2\min\{P(S \geq s_{\text{obs}}), P(S \leq s_{\text{obs}})\}$, using the symmetry of $B(n, \frac{1}{2})$.

Proposition 2.2 (Rejection region)

The smallest c with $P(S \geq c) \leq \alpha$ under H_0 is $c = Q_B(1-\alpha) + 1$, where Q_B is the quantile function of $B(n, \frac{1}{2})$.

Proof. $P(S \geq c) \leq \alpha \iff F_B(c-1) \geq 1 - \alpha$, and the smallest integer $c-1$ with this property is $Q_B(1-\alpha)$ (Proposition 1.1). ■

Because S is discrete, the attained size is usually below α. For large n, $(S - n/2)/\sqrt{n/4} \dot{\sim} N(0,1)$.

Ties. Observations equal to θ_0 are discarded and n is reduced accordingly; conditionally on the number discarded, the remaining signs are again i.i.d. fair coins whenever $P(X < \theta_0) = P(X > \theta_0)$.

Example 2.1

Twelve paired differences (after − before) have 9 positive signs. For $H_1 : \theta > 0$, $p = P(S \geq 9) = (220 + 66 + 12 + 1)/4096 = 0.073$.

Power. If the alternative population is known, $S \sim B(n, p_1)$ with $p_1 = P(X > \theta_0)$, so the power is $1 - F_B(c-1; n, p_1)$. It depends on the shape of F, not only on θ.

Example 2.2 (Sign test vs. z-test)

Let $n = 20$, $\theta_0 = 0$, $\theta = 0.5$, standard deviation 1. The size-≤ 0.05 sign test rejects when $S \geq 15$ (actual size 0.021).

- Normal $N(0.5, 1)$: $p_1 = \Phi(0.5) = 0.69$, power 0.38; the z-test has power $\Phi(0.5\sqrt{20} - 1.645) = 0.72$.
- Laplace with the same mean and variance (scale $1/\sqrt{2}$): $p_1 = 1 - \frac{1}{2}e^{-0.5\sqrt{2}} = 0.75$, power 0.63. The z-test still has power about 0.72.

The sign test gains ground when the tails are heavy. The asymptotic comparison is made precise in Section 2.3.

2.1.1 Confidence interval for the median

Proposition 2.3

Let F be continuous, $X_{(1)} < \cdots < X_{(n)}$ the order statistics and $l = Q_B(\alpha/2)$, $u = n - l + 1$. Then

$$P(X_{(l)} \leq \theta < X_{(u)}) = \sum_{k=l}^{n-l} \binom{n}{k} 2^{-n} \geq 1 - \alpha.$$

Proof. $X_{(l)} \leq \theta$ iff at most $n-l$ observations exceed θ, and $\theta < X_{(u)}$ iff at least $n-u+1 = l$ do. So the event is $\{l \leq S(\theta) \leq n-l\}$ with $S(\theta) \sim B(n, \frac{1}{2})$. By symmetry $P(S \leq l-1) = P(S \geq n-l+1)$, and each is $\leq \alpha/2$ by the choice of l (if $P(S \leq Q_B(\alpha/2)) = \alpha/2$ exactly, take $l = Q_B(\alpha/2) + 1$). ■

For $n = 15$ and $\alpha = 0.05$: $l = 4$, and $(X_{(4)}, X_{(12)})$ has coverage 0.965. For large n, the normal approximation $S \dot{\sim} N(n/2, n/4)$ gives the orders

$$l \approx \frac{n}{2} - z_{\alpha/2}\frac{\sqrt{n}}{2}, \qquad u \approx \frac{n}{2} + z_{\alpha/2}\frac{\sqrt{n}}{2} + 1 \qquad \text{(rounded to integers)},$$

which still yields an asymptotically valid, distribution-free interval. As the level decreases the interval shrinks to the sample median, the estimator associated with the sign test.

The same argument handles any quantile $Q(p)$: $\#\{X_i \leq Q(p)\} \sim B(n, p)$, so $(X_{(l)}, X_{(u)})$ covers $Q(p)$ with probability $\sum_{k=l}^{u-1} \binom{n}{k} p^k (1-p)^{n-k}$.

2.2 The Wilcoxon signed-rank test

Motivation. The sign test throws away magnitudes. If the distribution is symmetric, the ranks of $|X_i|$ can be used too; the signed-rank statistic remains distribution-free and is almost as efficient as the t-test under normality.

Assume now that F is continuous and symmetric about θ. Let $D_i = X_i - \theta_0$, $V_i = I(D_i > 0)$, and let R_i be the rank of $|D_i|$ among $|D_1|, \ldots, |D_n|$.

Definition (Signed-rank statistic)

$SR_+ = \sum_{i=1}^{n} V_i R_i$, the sum of the ranks of the positive deviations.

Lemma 2.4

Under H_0 (symmetry about θ_0), $(V_1, \dots, V_n)$ are i.i.d. Bernoulli($\frac{1}{2}$) and independent of $(|D_1|, \dots, |D_n|)$. Consequently

$$SR_+ \overset{d}{=} \sum_{k=1}^{n} kB_k, \qquad B_k \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(\tfrac{1}{2}).$$

Proof. For a symmetric continuous D, $\operatorname{sgn}(D)$ and $|D|$ are independent with $P(D > 0) = \frac{1}{2}$: $P(D > 0, |D| \le t) = P(0 < D \le t) = \frac{1}{2}P(|D| \le t)$. Independence across i gives the first claim. Given the ranks, (R_i) is a permutation π of $(1, \dots, n)$ and $\sum_i V_i R_i = \sum_k kV_{\pi^{-1}(k)}$, with $(V_{\pi^{-1}(k)})_k$ again i.i.d. fair coins. ■

Proposition 2.5

Under H_0, the null distribution of SR_+ depends only on n, is symmetric about $n(n+1)/4$, and

$$E(SR_+) = \frac{n(n+1)}{4}, \qquad \operatorname{Var}(SR_+) = \frac{n(n+1)(2n+1)}{24}, \qquad \frac{SR_+ - E(SR_+)}{\sqrt{\operatorname{Var}(SR_+)}} \xrightarrow{d} N(0,1).$$

Proof [Supplement]. From Lemma 2.4, $E = \frac{1}{2}\sum k$ and $\operatorname{Var} = \frac{1}{4}\sum k^2$. Replacing B_k by $1 - B_k$ (same law) maps SR_+ to $\frac{n(n+1)}{2} - SR_+$, giving symmetry. For normality use Lyapunov's CLT with $U_k = kB_k$: $\sum_k E|U_k - k/2|^3 = \sum k^3/8 = O(n^4)$, while $s_n^3 = \operatorname{Var}(SR_+)^{3/2} \asymp n^{9/2}$, so the Lyapunov ratio is $O(n^{-1/2}) \to 0$. ■

The exact null distribution is obtained by enumerating the 2^n equally likely sign patterns of the ranks (the *sign-flip* or re-randomization principle). The p-values are $P(SR_+ \ge sr_{\text{obs}})$ against $\theta > \theta_0$ and $P(SR_+ \le sr_{\text{obs}})$ against $\theta < \theta_0$. For the two-sided alternative, the p-value is twice the smaller one, which is legitimate here because the null distribution is symmetric.

Example 2.3

Deviations -0.3, 0.8, 1.5, 2.2, 2.9 from θ_0. The absolute ranks are $1, \dots, 5$ and only rank 1 is negative, so $SR_+ = 2 + 3 + 4 + 5 = 14$. Among the 32 sign patterns, $SR_+ \ge 14$ occurs for "all positive" (15) and "only rank 1 negative" (14), so the one-sided p-value is $2/32 = 0.0625$. The sign test gives $P(S \ge 4) = 6/32 = 0.19$: the ranks carry more information.

Ties. Zero deviations are dropped. Tied $|D_i|$ receive mid-ranks. The null distribution is then the sign-flip distribution of the actual (mid-)ranks, which no longer has the form in Lemma 2.4 but is still exact conditionally. For the normal approximation use $\operatorname{Var}(SR_+) = \frac{1}{4}\sum_i R_i^2$.

2.2.1 Walsh averages, the Hodges–Lehmann estimator and intervals

The $N = n(n+1)/2$ **Walsh averages** are $W_{ij} = (X_i + X_j)/2$, $1 \le i \le j \le n$.

Proposition 2.6

If no $|X_k - \theta|$ are tied and no W_{ij} equals θ, then $SR_+(\theta) = \#\{(i \le j) : W_{ij} > \theta\}$.

Proof [Supplement]. Let $d_k = X_k - \theta$, so $W_{ij} > \theta \iff d_i + d_j > 0$. Assign each pair $\{i, j\}$ to the index $k \in \{i, j\}$ with larger $|d_k|$. The sign of $d_i + d_j$ is the sign of d_k. Index k receives exactly the pairs $\{k, m\}$ with $|d_m| \le |d_k|$, i.e. R_k pairs. Hence $\#\{W_{ij} > \theta\} = \sum_{k:d_k>0} R_k$. ■

So the signed-rank statistic is the sign statistic applied to the Walsh averages, and Proposition 2.3 transfers.

Corollary 2.7

Let $W_{(1)} < \cdots < W_{(N)}$ and $l = Q_{SR}(\alpha/2)$ (quantile of the null distribution of SR_+), $u = N - l + 1$. Then $P(W_{(l)} \le \theta < W_{(u)}) \ge 1 - \alpha$. The point estimator obtained as the level shrinks is the **Hodges–Lehmann estimator** $\hat{\theta}_{HL} = \text{median}_{i \le j} W_{ij}$.

Proof. $\theta < W_{(a)} \iff SR_+(\theta) > N - a$, and $\theta \ge W_{(a)} \iff SR_+(\theta) \le N - a$. At the true θ, $SR_+(\theta)$ has the null distribution, which is symmetric about $N/2$. Repeat the proof of Proposition 2.3. ■

Paired data are handled by applying these one-sample methods to the differences.

2.3 Comparing the t-, sign and signed-rank procedures

Validity. The sign test is exact for any continuous F, and the signed-rank test for any continuous symmetric F. The t-test is exact only under normality. It is approximately valid when the CLT applies, but can be badly conservative for very heavy tails (where $\sigma^2 = \infty$, e.g. Cauchy or slash) and anticonservative for skewed F in moderate samples. For asymmetric F, mean and median differ, so the t-test does not even test the median. Typical behavior of the nominal 5% one-sided t-test:

- light tails (uniform): size slightly above 5% for $n \approx 5$, essentially correct by $n \approx 10$;
- infinite variance (slash, Cauchy): size well below 5% (very conservative), and this does *not* improve as n grows, because the CLT fails;
- skewed (exponential, testing the mean): convergence is slow, and the size can still be noticeably above 5% at $n = 100$; a corrected critical value is needed to restore it;
- skewed, testing the median with the t-statistic: the statistic is centered at the wrong value and the test is meaningless.

A test whose size exceeds α cannot be compared on power. Valid tests (or tests with corrected critical values) are compared by power at plausible alternatives, which depends on F, n and α.

Efficiency. The relative efficiency of estimators is $RE(\hat{\theta}_1, \hat{\theta}_2) = \text{MSE}(\hat{\theta}_2)/\text{MSE}(\hat{\theta}_1)$, and the **asymptotic relative efficiency** (ARE) is its limit as $n \to \infty$. For tests, the (Pitman) ARE is the limiting ratio N_2/N_1 of the sample sizes needed for equal power at alternatives $\theta_n \to \theta_0$. For these three procedures it equals the ARE of the associated estimators $\bar{X}_n$, median X_i and $\hat{\theta}_{HL}$.

For confidence intervals with the same (exact or asymptotic) coverage, define the efficiency as $1/\text{length}^2$. If $\hat{\theta} \dot{\sim} N(\theta, v/n)$, the interval $\hat{\theta} \pm z_{\alpha/2}\sqrt{v/n}$ has length

$$2z_{\alpha/2}\sqrt{v/n} \approx 2z_{\alpha/2}\sqrt{\text{MSE}(\hat{\theta})}.$$

Hence the ARE of two such intervals is the limit of the ratio of the mean squared errors of the two estimators, i.e. the ARE of the estimators. So estimators, tests and intervals are ranked the same way. For a normal population, e.g., $ARE(\bar{X}_n, \text{median } X_i) = \pi/2 \approx 1.57$.

Proposition 2.8

Let F have density f symmetric about θ, with $f(\theta) > 0$ and variance σ^2 (possibly infinite). Then

$$\sqrt{n}(\bar{X}_n - \theta) \xrightarrow{d} N(0, \sigma^2), \qquad \sqrt{n}(\text{median } X_i - \theta) \xrightarrow{d} N\Big(0, \frac{1}{4f(\theta)^2}\Big),$$
$$\sqrt{n}(\hat{\theta}_{HL} - \theta) \xrightarrow{d} N\Big(0, \frac{1}{12(\int f^2)^2}\Big).$$

Hence

$$ARE(\text{sign}, t) = 4\sigma^2 f(\theta)^2, \qquad ARE(SR, t) = 12\sigma^2\Big(\int f^2\Big)^2, \qquad ARE(\text{sign}, SR) = \frac{f(\theta)^2}{3(\int f^2)^2}.$$

Proof for the median; sketch for HL. $P(\text{median } X_i \le \theta + t/\sqrt{n}) = P(\#\{X_i \le \theta + t/\sqrt{n}\} \ge \lceil n/2 \rceil)$. The count is $B(n, p_n)$ with $p_n = \frac{1}{2} + f(\theta)t/\sqrt{n} + o(n^{-1/2})$. The normal approximation of the binomial gives the limit $\Phi(2f(\theta)t)$, i.e. variance $1/(4f(\theta)^2)$. For $\hat{\theta}_{HL}$ argue the same way with Proposition 2.6: $\hat{\theta}_{HL} \le \theta + t/\sqrt{n}$ iff $SR_+(\theta + t/\sqrt{n}) \le N/2$. Under the true θ, $SR_+(\theta + t/\sqrt{n})$ has mean $N/2 - n^{3/2}t\int f^2 + o(n^{3/2})$ and variance $\sim n^3/12$, giving the limit $\Phi(t\sqrt{12}\int f^2)$. ■

F	ARE(sign, t)	ARE(SR, t)	ARE(sign, SR)
Uniform	$1/3$	1	$1/3$
Normal	$2/\pi \approx 0.64$	$3/\pi \approx 0.955$	$2/3$
Logistic	$\pi^2/12 \approx 0.82$	$\pi^2/9 \approx 1.10$	$3/4$
Laplace	2	1.5	$4/3$
Cauchy, slash	∞	∞	finite

(Check for $N(0,1)$: $f(0)^2 = \frac{1}{2\pi}$, $\int \phi^2 = \frac{1}{2\sqrt{\pi}}$. For Laplace $\frac{1}{2}e^{-|x|}$: $\sigma^2 = 2$, $f(0) = \frac{1}{2}$, $\int f^2 = \frac{1}{4}$.) Over all symmetric F, $ARE(SR, t) \ge 0.864$ (Hodges–Lehmann). So the signed-rank test costs little under normality and can gain a lot under heavy tails. The sign test wins only for very heavy tails.

Chapter 3. Permutation Tests

3.1 Two sources of randomness

- **Randomization model.** N fixed units are randomly assigned, m to treatment 1 and $n = N - m$ to treatment 2. The responses are fixed numbers; only the assignment is random. Inference concerns these N units.
- **Population model.** $X_1, \ldots, X_m \overset{\text{i.i.d.}}{\sim} F$ and $Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} G$ are random samples. Inference concerns F and G, e.g. $H_0 : F = G$.

Both lead to the same computation: the *permutation* (or *randomization*) distribution of a statistic over all re-assignments of the observed values.

3.2 The randomization test

Under H_0: "treatment has no effect," each unit's response would be the same under either treatment. The observed assignment is one of $\binom{N}{m}$ equally likely assignments, and every other assignment would have produced the same pooled set of responses. For a statistic D (e.g. $\bar{Y} - \bar{X}$), let $d_1, \ldots, d_K$, $K = \binom{N}{m}$, be its values over all assignments. The upper-tailed p-value is

$$p = \frac{1}{K} \sum_{k=1}^{K} I(d_k \geq D_{\text{obs}}). \tag{3.1}$$

For a two-sided test use $|d_k| \geq |D_{\text{obs}}|$ (or a symmetric criterion), not twice the smaller one-sided value: the randomization distribution need not be symmetric.

Example 3.1

Five patients: two random patients get a new treatment with recovery times $3, 5$; the others have $8, 9, 12$. Let $D = \bar{Y}_{\text{old}} - \bar{X}_{\text{new}}$. The pooled total is 37, so for a treated pair with sum s, $D = (37 - s)/3 - s/2$, a decreasing function of s. The $\binom{5}{2} = 10$ pair sums are $8, 11, 12, 13, 14, 15, 17, 17, 20, 21$. The observed sum 8 is the smallest, so $D_{\text{obs}} = 5.67$ is the largest value and $p = 1/10$.

Proposition 3.1 (Exactness)

Under H_0, if the d_k have no ties and $\alpha = j/K$, the randomization test rejecting when $p \leq \alpha$ has type I error exactly α. With ties it is at most α.

Proof. Under H_0 the multiset $\{d_k\}$ does not depend on which assignment occurred, and the realized assignment is uniform. So D_{obs} is a uniformly chosen element of the multiset. Without ties, $p \leq j/K$ iff D_{obs} is among the j largest values, which has probability j/K. With ties, p only gets larger. ■

Proposition 3.2 (Equivalent statistics)

Over all re-assignments of a fixed pooled sample, $\sum_j Y_j$, $\bar{Y} - \bar{X}$ and the pooled two-sample t-statistic are strictly increasing functions of each other, hence give identical permutation p-values.

Proof. The pooled total $\sum X_i + \sum Y_j$ is fixed, so $\bar{Y} - \bar{X} = \frac{N}{mn}\sum Y_j - \frac{\text{total}}{m}$ is increasing in $\sum Y_j$. The total sum of squares SST is also fixed, and $SST = (N-2)s_p^2 + \frac{mn}{N}D^2$ with $D = \bar{Y} - \bar{X}$. Hence

$$T = \frac{D}{\sqrt{\frac{1}{m}+\frac{1}{n}}\sqrt{(SST - \frac{mn}{N}D^2)/(N-2)}} = g(D), \qquad g'(D) \propto \frac{SST}{(SST - \frac{mn}{N}D^2)^{3/2}} > 0.$$

■

Statistics such as a difference of medians or of trimmed means are *not* monotone in D and can give different (often more powerful, for skewed or heavy-tailed data) tests.

3.3 Permutation tests in the population model

Motivation. Randomization justifies permutation tests in designed experiments. The same tests remain valid for random samples from two populations, because under H_0 the pooled sample is exchangeable.

Definition (Exchangeability)

$(Z_1, \ldots, Z_N)$ is **exchangeable** if $(Z_{\pi(1)}, \ldots, Z_{\pi(N)}) \overset{d}{=} (Z_1, \ldots, Z_N)$ for every permutation π.

I.i.d. variables are exchangeable. So are equicorrelated normal variables $N_N(\mu\mathbf{1}, \sigma^2[(1-\rho)I + \rho\mathbf{1}\mathbf{1}'])$, which are not independent. Two independent samples from $N(\mu, 1)$ and $N(\mu, 4)$ are *not* exchangeable, even though their means agree.

Theorem 3.3 (Validity of permutation tests)

Let $Z = (Z_1, \ldots, Z_N)$ be exchangeable under H_0, let $\mathcal{G}$ be a group of permutations, and let $p(Z) = \frac{1}{|\mathcal{G}|}\sum_{g\in\mathcal{G}} I(T(gZ) \geq T(Z))$. Then $P_{H_0}(p(Z) \leq \alpha) \leq \alpha$ for every α, with equality when $\alpha|\mathcal{G}|$ is an integer and there are no ties. The statement holds conditionally on the multiset of observed values, hence also unconditionally.

Proof. Let U be uniform on $\mathcal{G}$, independent of Z. By exchangeability $UZ \overset{d}{=} Z$. Since $\mathcal{G}$ is a group, the orbit $\{gUZ : g \in \mathcal{G}\}$ equals $\{gZ\}$, so $p(UZ)$ is computed from the same multiset of values $\{T(gZ)\}$, and $T(UZ)$ is a uniformly chosen member of it. A uniformly chosen member lies among the top j (counting ties generously) with probability at most $j/|\mathcal{G}|$. Hence $P(p(Z) \leq \alpha) = P(p(UZ) \leq \alpha) \leq \alpha$. Taking expectations over the conditioning gives the unconditional statement. ■

For two samples under $H_0 : F = G$ the pooled observations are i.i.d., hence exchangeable, and the relevant re-assignments are the $\binom{N}{m}$ splits (each realized by $m!\,n!$ permutations). The permutation test is **conditional inference**: the observed values are held fixed and only their labels are random.

3.4 Random (Monte Carlo) permutation tests

When $\binom{N}{m}$ is too large, draw M random relabelings, compute $T_1, \ldots, T_M$, set $T_0 = T_{\text{obs}}$, and use

$$\hat{p} = \frac{1 + \sum_{i=1}^{M} I(T_i \geq T_{\text{obs}})}{M+1}. \tag{3.2}$$

Proposition 3.4

Under H_0, $P(\hat{p} \le \alpha) \le \alpha$ for every M. Moreover $\hat{p}$ estimates the exact permutation p-value with standard error about $\sqrt{p(1-p)/M}$.

Proof. Given the multiset of values, $T_0, T_1, \ldots, T_M$ are exchangeable under H_0: the observed labeling is itself a uniform draw, like the M random ones. The rank of T_0 among the $M+1$ values is therefore uniform (with ties broken against rejection), and $\hat{p} \le j/(M+1)$ iff T_0 is among the j largest. For the second claim, $\hat{p} \approx$ the mean of M Bernoulli(p) indicators. ■

This is why the "+1" is used: it makes the Monte Carlo test exactly valid, not just approximately.

Generating a random relabeling. Draw a uniform random permutation of $1, \ldots, N$, and give group 1 either to the units in its first m positions or to the units that receive the first m labels. Either way, each m-subset arises from exactly $m!\,n!$ of the $N!$ permutations, so every split has probability $1/\binom{N}{m}$.

Two-sample alternatives and p-values. With $H_0 : F = G$, the one-sided alternatives are stochastic orderings, $G(x) \le F(x)$ for all x (Y tends to be larger) or the reverse. Under a shift model they become $\Delta > 0$ or $\Delta < 0$. For $D = \bar{Y} - \bar{X}$ with values $d_0 = D_{\text{obs}}, d_1, \ldots, d_M$:

$$p_{\text{upper}} = \frac{\#\{d_i \ge d_0\}}{M+1}, \qquad p_{\text{lower}} = \frac{\#\{d_i \le d_0\}}{M+1}, \qquad p_{\text{two}} = \frac{\#\{|d_i| \ge |d_0|\}}{M+1}.$$

With $M = 1999$, the standard error at $p = 0.05$ is about 0.005.

3.5 Other permutation tests

One sample: sign flipping. If $X_i - \theta_0$ are independent and symmetric about 0 under H_0, then $(\pm(X_1-\theta_0), \ldots, \pm(X_n-\theta_0))$ have the same distribution for all 2^n sign choices. The permutation test over sign flips with statistic $\sum I(X_i > \theta_0)$ is the sign test. With $\sum V_i R_i$ it is the signed-rank test. With the one-sample t-statistic it is a symmetric-F version of the t-test (so the tests of Chapter 2 are permutation tests).

2×2 tables: Fisher's exact test. With two binary variables (e.g. exposure and disease) and H_0: independence (or equal proportions), hold one variable fixed and permute the other. With margins fixed, the count D in the (exposed, diseased) cell follows the hypergeometric distribution:

$$P(D = d) = \binom{m}{d}\binom{N-m}{k-d}\Big/\binom{N}{k},$$

where m is the number diseased and k the number exposed among N.

Example 3.2

$N = 20$ subjects, $m = 10$ diseased, $k = 6$ exposed, and 5 of the exposed are diseased. Then $P(D \ge 5) = [\binom{10}{5}\binom{10}{1} + \binom{10}{6}\binom{10}{0}]/\binom{20}{6} = (2520 + 210)/38760 = 0.070$.

Summary. Permutation tests (i) need only exchangeability under H_0, (ii) allow any statistic, and (iii) give exact p-values by computation. Their limits: they test only hypotheses that make the data exchangeable (e.g. not "equal means, unequal variances"), and interval estimation requires inverting many tests.

Chapter 4. Inference for Two Populations

4.1 Models for two populations

$X_1, \dots, X_m \overset{\text{i.i.d.}}{\sim} F$ and $Y_1, \dots, Y_n \overset{\text{i.i.d.}}{\sim} G$ independent, $N = m + n$. Typical models:

- **Shift:** $G(x) = F(x - \Delta)$, i.e. $Y \overset{d}{=} X + \Delta$.
- **Scale:** $G(x) = F(x/\gamma)$ (common center 0), or location–scale $G(x) = F((x - \Delta)/\gamma)$.
- **General:** $H_0 : F = G$ against $F \neq G$.

Permutation tests require exchangeability under H_0, which holds when H_0 forces $F = G$: the shift model with $\Delta = 0$, the scale model with $\gamma = 1$, or the general null. It fails for "equal locations, different scales." For that problem see Section 4.6.

For normal data the pooled t-test is optimal when variances are equal, and Welch's test, with $T = (\bar{Y} - \bar{X})/\sqrt{s_X^2/m + s_Y^2/n}$ and Satterthwaite degrees of freedom, is approximately valid when they are not.

4.2 The Wilcoxon rank-sum (Mann–Whitney) test

Motivation. Replacing observations by their ranks in the pooled sample makes the permutation distribution independent of the data, so it can be tabulated once. The rank-sum statistic also estimates $P(X < Y)$, which gives it a natural interpretation.

Rank all N observations together. Let $R_1, \dots, R_n$ be the ranks of the Y's.

Definition (Rank-sum and Mann–Whitney statistics)

$W = \sum_{j=1}^{n} R_j$ and $U = \sum_{i=1}^{m} \sum_{j=1}^{n} I(Y_j > X_i)$ (with $\frac{1}{2}$ for ties).

Proposition 4.1

Without ties, $W = U + n(n+1)/2$.

Proof. $R_j = 1 + \#\{k : Y_k < Y_j\} + \#\{i : X_i < Y_j\}$. Summing over j, the middle terms give $0 + 1 + \cdots + (n-1)$. Hence $W = n + \frac{n(n-1)}{2} + U$. ■

Under $H_0 : F = G$ (continuous), the rank vector of the Y's is a uniformly random n-subset of $\{1, \dots, N\}$. So the null distribution of W depends only on (m, n), and the test is distribution-free. It is the permutation test applied to ranks; large W indicates $\Delta > 0$.

Example 4.1

$X = \{1.1, 2.4\}$, $Y = \{2.0, 3.5, 4.2\}$. The pooled ranks of Y are $2, 4, 5$, so $W = 11$ and $U = 5$. Among the $\binom{5}{3} = 10$ rank subsets, only $\{2, 4, 5\}$ and $\{3, 4, 5\}$ have sum ≥ 11, so $P(W \geq 11) = 0.2$.

4.2.1 Large-sample approximation via sampling without replacement

Theorem 4.2

Let $\bar{X}_n$ be the mean of a simple random sample of size n drawn without replacement from a finite population $\{a_1, \dots, a_N\}$ with mean μ and variance $\sigma^2 = \frac{1}{N}\sum(a_k - \mu)^2$. Then

$$E\bar{X}_n = \mu, \qquad \operatorname{Var}\bar{X}_n = \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1}.$$

Proof [Supplement]. Write $n\bar{X}_n = \sum_k Z_k a_k$ with inclusion indicators Z_k, $P(Z_k = 1) = n/N$ and $P(Z_k = Z_l = 1) = \frac{n(n-1)}{N(N-1)}$ for $k \neq l$. So $\operatorname{Var} Z_k = \frac{n}{N}(1 - \frac{n}{N})$ and $\operatorname{Cov}(Z_k, Z_l) = -\frac{n(N-n)}{N^2(N-1)}$. Take $\mu = 0$ without loss, so $\sum_k a_k^2 = N\sigma^2$ and $\sum_{k \neq l} a_k a_l = -N\sigma^2$. Then

$$\operatorname{Var}\Big(\sum Z_k a_k\Big) = N\sigma^2 \frac{n(N-n)}{N^2} + N\sigma^2 \frac{n(N-n)}{N^2(N-1)} = n\sigma^2 \frac{N-n}{N-1}.$$

■

The factor $\frac{N-n}{N-1}$ is the **finite-population correction**. A CLT holds when $\min(n, N-n) \to \infty$ and no single a_k dominates, e.g. $\max_k(a_k - \mu)^2/(\min(n, N-n)\sigma^2) \to 0$ (Hájek). For ranks, $\{a_k\} = \{1, \dots, N\}$ has $\mu = \frac{N+1}{2}$ and $\sigma^2 = \frac{N^2-1}{12}$, so

$$E(W) = \frac{n(N+1)}{2}, \qquad \operatorname{Var}(W) = \frac{mn(N+1)}{12}, \qquad \frac{W - E(W)}{\sqrt{\operatorname{Var}(W)}} \xrightarrow{d} N(0,1).$$

With ties, use the mid-ranks as the finite population: the mean is unchanged, and the variance is recomputed from (4.1) below,

$$\operatorname{Var}(W) = n^2 \cdot \frac{\sigma_R^2}{n} \cdot \frac{N-n}{N-1}, \qquad \sigma_R^2 = \text{population variance of the mid-ranks.} \tag{4.1}$$

The same theorem gives the large-sample permutation distribution of $\bar{Y}$ itself (population = the pooled observations), which is why permutation and normal-theory p-values agree for large samples.

4.2.2 Estimating the shift

In the shift model, $U(\Delta_0) = \#\{(i,j) : Y_j - X_i > \Delta_0\}$ is the Mann–Whitney statistic of X and $Y - \Delta_0$. At the true Δ it has the null distribution, which is symmetric about $mn/2$.

Proposition 4.3

Let $D_{(1)} \le \cdots \le D_{(mn)}$ be the ordered differences $Y_j - X_i$, $k_l = Q_U(\alpha/2)$ and $k_u = mn - k_l + 1$. Then $P(D_{(k_l)} \le \Delta < D_{(k_u)}) = P(k_l \le U \le k_u - 1) \ge 1 - \alpha$. The associated point estimator is the Hodges–Lehmann shift estimator $\hat{\Delta} = \operatorname{median}_{i,j}(Y_j - X_i)$.

Proof. $D_{(k_l)} \le \Delta \iff U(\Delta) \le mn - k_l$ and $\Delta < D_{(k_u)} \iff U(\Delta) \ge mn - k_u + 1$. Since $U(\Delta)$ has the null distribution, symmetric about $mn/2$, the probability equals $P(k_l \le U \le k_u - 1)$. ■

In Example 4.1 the six differences $Y_j - X_i$ are -0.4, 0.9, 1.1, 1.8, 2.4, 3.1, so $\hat{\Delta} = 1.45$. For large samples, $Q_U(\alpha) \approx \frac{mn}{2} - z_\alpha\sqrt{mn(N+1)/12}$.

Ranks as a probability transform. Under H_0, $U_i = F(Z_i)$ are i.i.d. uniform for all N pooled observations, and $E\,U_{(k)} = k/(N+1)$. So, up to the factor $N+1$, the ranks are the expected uniform order statistics: the rank-sum test is a permutation test applied after (implicitly) transforming the unknown F to the uniform. This is why its null distribution is the same for every continuous F, and also why it needs $F = G$ under H_0: if the null allows unequal shapes, the transform is not common to both samples.

Paired data. With pairs (X_i, Y_i), analyze the differences $Y_i - X_i$ with the one-sample methods of Chapter 2.

4.3 Tests for scale

Assume a common center and $G(x) = F(x/\gamma)$. Ranks alone are useless here (a scale change does not shift W), so the ranks are replaced by *scores* that are large in the tails:

- **Ansari–Bradley:** scores $1, 2, 3, \ldots$ from both ends inward. A small sum of Y-scores means Y is more dispersed.
- **Siegel–Tukey:** an alternating assignment of $1, 2, \ldots, N$ from both ends, so the null distribution equals that of W (historically convenient, now unnecessary).
- **Mood:** scores $(R - \frac{N+1}{2})^2$. A large sum of Y-scores means Y is more dispersed.

Each is a permutation test of the score sum, so it is distribution-free under $F = G$.

If the centers may differ, other statistics are available.

- Under normality, use $F = S_Y^2/S_X^2 \sim F_{n-1,m-1}$ when $\gamma = 1$. This test is very sensitive to non-normality and outliers.
- The ratio of median absolute deviations $\mathrm{MAD}(Y)/\mathrm{MAD}(X)$ is a robust alternative.
- Alternatively, use any of the score statistics after centering each sample at its own median.

With unequal centers $F = G$ is no longer the null, so for the last two statistics the null distribution is obtained by permuting the median-centered observations (mean-centered for the F ratio). This is only *approximately* valid, since the centered observations are only approximately exchangeable. Without centering, a location difference masquerades as a scale difference.

4.4 The Kolmogorov–Smirnov test

For general $H_0 : F = G$ against $F \neq G$, compare the empirical distribution functions $\hat{F}_m(x) = \frac{1}{m}\sum I(X_i \le x)$ and $\hat{G}_n$:

$$K = \sqrt{\frac{mn}{m+n}}\,\sup_x |\hat{F}_m(x) - \hat{G}_n(x)|.$$

The supremum is attained at an observation, so K is a function of the ranks only. Hence its null distribution is distribution-free (for continuous F). As $m, n \to \infty$, $P(K \le x) \to 1 - 2\sum_{j\ge1}(-1)^{j-1}e^{-2j^2x^2}$ (Kolmogorov). The KS test detects any difference (location, scale, shape), but against a pure shift it is less powerful than the rank-sum test.

Why K depends only on ranks [Supplement]. Between consecutive pooled order statistics both empirical distribution functions are constant, so the supremum is a maximum over the N pooled points. At the kth smallest pooled point, $\hat{F}_m$ and $\hat{G}_n$ equal the proportions of X's and Y's among the k smallest, which are determined by the ranks. Under H_0 the rank configuration is uniform over $\binom{N}{m}$ splits. ■

4.5 Which test?

In the shift model with $F = G$ under H_0, every permutation or rank test is exact. Pitman AREs of the rank-sum test relative to the t-test equal $12\sigma^2(\int f^2)^2$, as in the one-sample case:

F	uniform	normal	logistic	Laplace	Cauchy
ARE(Wilcoxon, t)	1	$3/\pi \approx 0.955$	$\pi^2/9 \approx 1.10$	1.5	∞

(For the exponential shift family the value is 3.) On type I error: rank and permutation tests are exact. The two-sample t-test is close to nominal for light tails and conservative for heavy-tailed or skewed populations. In small-sample power studies, the permutation t-test is slightly ahead for uniform data, the t-test for normal data, the rank-sum test for Laplace-type tails, and the difference-of-medians permutation test for Cauchy-type tails. Roughly: the t-test or the permutation t-test for light tails and small samples; the rank-sum test for moderate to heavy tails; the difference-of-medians permutation test for extremely heavy tails.

4.6 Unequal variances under the null

If H_0 is "equal locations" but the scales differ, neither the rank-sum test nor any permutation test is guaranteed to hold its level. Options are Welch's t-test (CLT-based), a variance-stabilizing transformation, or a studentized rank statistic (Fligner–Policello, Chung–Romano) that is asymptotically valid. The bootstrap tests of Chapter 9 also apply.

Chapter 5. Association and Independence of Two Variables

5.1 Pearson correlation

For a pair (X, Y), $\rho = \mathrm{Cov}(X, Y)/\sqrt{\mathrm{Var}\, X\, \mathrm{Var}\, Y} \in [-1, 1]$ measures *linear* association. Independence implies $\rho = 0$; the converse holds under bivariate normality but not in general (e.g. $Y = X^2$ with X symmetric gives $\rho = 0$). From pairs (X_i, Y_i), $i = 1, \ldots, n$, the sample correlation is $r = S_{XY}/\sqrt{S_{XX}S_{YY}}$. The least squares slope is $\hat{\beta} = S_{XY}/S_{XX} = r\sqrt{S_{YY}/S_{XX}}$. For the population, $\beta = \mathrm{Cov}(X, Y)/\mathrm{Var}\, X = \rho\, \sigma_Y/\sigma_X$, so testing $\beta = 0$ in a simple regression and testing $\rho = 0$ are the same problem.

Proposition 5.1

If (X_i, Y_i) are i.i.d. bivariate normal with $\rho = 0$, then $T = r\sqrt{(n-2)/(1-r^2)} \sim t_{n-2}$.

Proof. Condition on the X's. Then Y_i are i.i.d. normal, i.e. the regression model with $\beta = 0$, and the slope t-statistic $\hat{\beta}/\widehat{\mathrm{se}}(\hat{\beta}) \sim t_{n-2}$. With $SSE = S_{YY}(1 - r^2)$ and $\widehat{\mathrm{se}}^2 = SSE/((n-2)S_{XX})$, one gets $\hat{\beta}/\widehat{\mathrm{se}} = r\sqrt{(n-2)/(1-r^2)}$. The conditional law does not depend on the X's. ■

Permutation test of independence. Under H_0: $X \perp Y$, the Y's can be permuted among the fixed X's without changing the joint law, so shuffling Y gives an exact test with any statistic, e.g. r. The alternative it is sensitive to is linear association. Note that H_0 is independence, not $\rho = 0$: if $\rho = 0$ but X, Y are dependent, shuffling is not justified.

Proposition 5.2

Over all $n!$ permutations of $(Y_1, \ldots, Y_n)$ with the X's fixed, the permutation distribution of r has mean 0 and variance $1/(n-1)$, whatever the data.

Proof. Let $a_i = X_i - \bar{X}$, $b_i = Y_i - \bar{Y}$ and π uniform. Then $r = \sum_i a_i b_{\pi(i)}/\sqrt{S_{XX}S_{YY}}$ and $Eb_{\pi(i)} = 0$. Also $\mathrm{Var}\, b_{\pi(i)} = S_{YY}/n$ and $\mathrm{Cov}(b_{\pi(i)}, b_{\pi(j)}) = -S_{YY}/(n(n-1))$ for $i \neq j$. With $\sum_{i \neq j} a_i a_j = -S_{XX}$,

$$\mathrm{Var}\left(\sum a_i b_{\pi(i)}\right) = \frac{S_{XX}S_{YY}}{n} + \frac{S_{XX}S_{YY}}{n(n-1)} = \frac{S_{XX}S_{YY}}{n-1}.$$

■

Combined with a finite-population CLT, $\sqrt{n-1}\, r$ is approximately $N(0, 1)$ under the permutation distribution, a distribution-free counterpart of Proposition 5.1.

5.2 Spearman's rank correlation

Replace X_i, Y_i by their ranks R_i, S_i and compute the Pearson correlation: $r_s = \mathrm{Corr}(R, S)$. It measures *monotone* association, is invariant under increasing transformations of either variable, and is robust to outliers.

Proposition 5.3

Without ties, $r_s = 1 - \dfrac{6\sum_i (R_i - S_i)^2}{n(n^2-1)}$.

Proof. Both rank vectors are permutations of $1, \dots, n$, with mean $\frac{n+1}{2}$ and $\sum(R_i - \bar{R})^2 = \frac{n(n^2-1)}{12}$. Since $\sum(R_i - S_i)^2 = \sum(R_i - \bar{R})^2 + \sum(S_i - \bar{S})^2 - 2\sum(R_i - \bar{R})(S_i - \bar{S})$, we get $\sum(R_i - \bar{R})(S_i - \bar{S}) = \frac{n(n^2-1)}{12} - \frac{1}{2}\sum(R_i - S_i)^2$. Divide by $\frac{n(n^2-1)}{12}$. ■

The permutation test with r_s tests independence against monotone association. It is distribution-free, and Proposition 5.2 applies to it verbatim.

5.3 Kendall's tau

Motivation. Pearson's correlation measures linear association and is sensitive to outliers. Kendall's tau counts concordant minus discordant pairs, so it measures monotone association and depends only on ranks.

A pair of observations $i < j$ is **concordant** if $(X_i - X_j)(Y_i - Y_j) > 0$ and **discordant** if < 0. Let $U_{ij} = \operatorname{sgn}[(X_i - X_j)(Y_i - Y_j)]$.

Definition (Kendall's tau)

Sample: $\tau_n = \binom{n}{2}^{-1} \sum_{i<j} U_{ij} = (N_c - N_d)/\binom{n}{2}$. Population: for independent copies $(X, Y), (X', Y')$, $\tau = P\{(X - X')(Y - Y') > 0\} - P\{(X - X')(Y - Y') < 0\}$.

τ_n is unbiased for τ. Unlike r_s, it has a clear population counterpart.

Example 5.1

$x = (1, 2, 3, 4, 5)$, $y = (2, 1, 4, 3, 5)$. Only the pairs $(1, 2)$ and $(3, 4)$ are discordant, so $\tau_n = (8 - 2)/10 = 0.6$. The rank differences are $(-1, 1, -1, 1, 0)$, so by Proposition 5.3, $r_s = 1 - 6 \cdot 4/120 = 0.8$. Pearson's r is also 0.8 here, since the data are their own ranks.

Proposition 5.4 (Normal case)

If (X, Y) is bivariate normal with correlation ρ, then $\tau = \frac{2}{\pi} \arcsin \rho$.

Proof [Supplement]. $(U, V) = (X - X', Y - Y')$ is centered normal with correlation ρ. Write $U = Z_1$ and $V = \rho Z_1 + \sqrt{1 - \rho^2} Z_2$, then pass to polar coordinates $(Z_1, Z_2) = R(\cos\Theta, \sin\Theta)$ with Θ uniform, and set $\rho = \sin\phi$. Then $V = R\sin(\Theta + \phi)$. The set where $\cos\Theta$ and $\sin(\Theta + \phi)$ share a sign has total angle $\pi + 2\phi$, so $P(UV > 0) = \frac{1}{2} + \frac{\phi}{\pi}$ and $\tau = 2P(UV > 0) - 1 = \frac{2\phi}{\pi}$. ■

This gives a robust estimator $\hat{\rho} = \sin(\pi\tau_n/2)$ of a normal correlation in the presence of outliers.

Proposition 5.5

If $X \perp Y$ with continuous margins, then $E\tau_n = 0$ and $\operatorname{Var}\tau_n = \dfrac{2(2n+5)}{9n(n-1)}$. Moreover $\tau_n / \sqrt{\operatorname{Var}\tau_n} \xrightarrow{d} N(0, 1)$.

Proof [Supplement]. Each $U_{ij} = \pm 1$ with probability $\frac{1}{2}$, so $\operatorname{Var} U_{ij} = 1$. Pairs with no common index give independent U's. For a shared index, independence of X and Y gives $E(U_{12}U_{13}) = \left(E[\operatorname{sgn}(X_1 - X_2)\operatorname{sgn}(X_1 - X_3)]\right)^2 = \left(E(2V - 1)^2\right)^2 = \frac{1}{9}$ with V uniform. There are $\binom{n}{2} \cdot 2(n-2)$ ordered pairs of pairs sharing one index, so

$$\operatorname{Var}\Big(\sum_{i<j} U_{ij}\Big) = \binom{n}{2}\Big(1 + \frac{2(n-2)}{9}\Big) = \binom{n}{2}\frac{2n+5}{9},$$

and dividing by $\binom{n}{2}^2$ gives the formula. Asymptotic normality follows from Hoeffding's theory of U-statistics. ■

Under independence the exact null distribution of τ_n is its permutation distribution (shuffle Y), which is distribution-free. Without ties this is Kendall's tabulated null distribution. So there are three equivalent ways to test independence:

(i) a (random) permutation test;

(ii) Kendall's exact null distribution;

(iii) the normal approximation of Proposition 5.5, already good for $n \approx 10$ except in the extreme tails. Pairs with $(X_i - X_j)(Y_i - Y_j) = 0$ give $U_{ij} = 0$. Software usually reports the tie-corrected version $\tau_b = (N_c - N_d)/\sqrt{(N_0 - T_X)(N_0 - T_Y)}$, where $N_0 = \binom{n}{2}$ and T_X, T_Y count the pairs tied in X or in Y. With ties only the permutation test remains exact.

Which measure? Under bivariate normality, use the t-test of r. With heavy-tailed margins but linear dependence, rank-based r_s and τ_n give more powerful tests. For monotone nonlinear dependence, r_s or τ_n are the natural measures. All three can be near zero for strong non-monotone dependence (e.g. $Y = \sin(3X) + \epsilon$), so failing to reject independence does not mean independence.

Chapter 6. Permutation Tests for Analysis of Variance

6.1 One-way layout

Motivation. To compare k groups without normality, rank all observations together and compare average ranks across groups; this is the Kruskal–Wallis test, the rank analogue of one-way ANOVA.

Observations $X_{ij} = \mu_i + \epsilon_{ij}$, $i = 1, \dots, k$, $j = 1, \dots, n_i$, $n = \sum n_i$, with i.i.d. errors. We test $H_0 : \mu_1 = \cdots = \mu_k$.

The F-test. With group means $\bar{X}_i$ and grand mean $\bar{X}$,

$$SS_{\mathrm{Tr}} = \sum_i n_i(\bar{X}_i - \bar{X})^2, \qquad SS_E = \sum_{i,j}(X_{ij} - \bar{X}_i)^2, \qquad F = \frac{SS_{\mathrm{Tr}}/(k-1)}{SS_E/(n-k)}.$$

Proposition 6.1

$SS_{\mathrm{Tot}} := \sum_{i,j}(X_{ij} - \bar{X})^2 = SS_{\mathrm{Tr}} + SS_E$. Under H_0 with normal errors, $F \sim F_{k-1,n-k}$.

Proof [Supplement]. Expand $X_{ij} - \bar{X} = (X_{ij} - \bar{X}_i) + (\bar{X}_i - \bar{X})$. The cross term is $2\sum_i(\bar{X}_i - \bar{X})\sum_j(X_{ij} - \bar{X}_i) = 0$. Under normality, $SS_E/\sigma^2 \sim \chi^2_{n-k}$ is independent of the group means. $SS_{\mathrm{Tr}}/\sigma^2$ is the squared length of the projection of the standard normal vector $(\sqrt{n_i}(\bar{X}_i - \mu)/\sigma)_i$ onto the complement of $(\sqrt{n_i})_i$, hence χ^2_{k-1}. ■

Permutation F-test. Under H_0 (equal distributions) all n observations are exchangeable, so relabel them among the groups: there are $n!/\prod n_i!$ equally likely splits. Since SS_{Tot} is fixed, $F = \frac{n-k}{k-1} \cdot \frac{SS_{\mathrm{Tr}}}{SS_{\mathrm{Tot}} - SS_{\mathrm{Tr}}}$ is increasing in SS_{Tr}, and both give the same test. For moderately large samples the permutation distribution of F is close to $F_{k-1,n-k}$ even without normality.

6.1.1 The Kruskal–Wallis test

Rank all observations jointly. Let $\bar{R}_i$ be the mean rank of group i and S_R^2 the sample variance of all ranks ($= n(n+1)/12$ without ties).

Definition (Kruskal–Wallis statistic)

$KW = \frac{1}{S_R^2}\sum_{i=1}^k n_i\Big(\bar{R}_i - \frac{n+1}{2}\Big)^2$, which without ties equals $\frac{12}{n(n+1)}\sum_i n_i(\bar{R}_i - \frac{n+1}{2})^2$.

Proposition 6.2

(i) The permutation test based on KW coincides with the permutation F-test applied to the ranks.
(ii) Under H_0, $KW \xrightarrow{d} \chi^2_{k-1}$ as all $n_i \to \infty$.

Proof. (i) For ranks, $SS^R_{\mathrm{Tot}} = (n-1)S_R^2$ is fixed and $SS^R_{\mathrm{Tr}} = S_R^2\, KW$, so $F_R = \frac{n-k}{k-1}\frac{KW}{n-1-KW}$ is increasing in KW.

(ii) With $Z_i = \sqrt{n_i}(\bar{R}_i - \bar{R})/S_R$, the without-replacement covariances (Theorem 4.2) give $\mathrm{Cov}(Z) \to I - \mathbf{u}\mathbf{u}'$ with $\mathbf{u} = (\sqrt{n_i/n})_i$, a rank-$(k-1)$ projection. A multivariate finite-population CLT makes Z asymptotically $N(0, I - \mathbf{u}\mathbf{u}')$, so $KW = \|Z\|^2 \xrightarrow{d} \chi^2_{k-1}$. ■

In small samples the χ^2 approximation can be poor in the tails, so the exact or Monte Carlo permutation distribution is preferred.

6.1.2 Ordered alternatives

Against $H_1 : \mu_1 \le \mu_2 \le \cdots \le \mu_k$ (at least one strict), the **Jonckheere–Terpstra** statistic sums the pairwise Mann–Whitney counts:

$$JT = \sum_{i<l} U_{il}, \qquad U_{il} = \#\{(a,b) : X_{ia} < X_{lb}\}.$$

Large JT supports the trend. It equals (up to an affine map) Kendall's τ between the observations and the group index, and its permutation distribution gives an exact test. Using the order information makes it more powerful than KW against monotone trends.

6.2 Randomized complete block design

k treatments in each of b blocks, one observation per cell: $X_{ij} = \mu + t_i + \beta_j + \epsilon_{ij}$. We test $H_0 : t_1 = \cdots = t_k$.

Classical F. $SS_{\text{Tr}} = b\sum_i(\bar X_{i\cdot} - \bar X)^2$ and $SS_E = \sum_{i,j}(X_{ij} - \bar X_{i\cdot} - \bar X_{\cdot j} + \bar X)^2$. Under normality, $F = \frac{SS_{\text{Tr}}/(k-1)}{SS_E/((k-1)(b-1))} \sim F_{k-1,(k-1)(b-1)}$.

Why shuffle within blocks. Under H_0, $X_{ij} = \mu + \beta_j + \epsilon_{ij}$: observations in the same block are exchangeable, but observations in different blocks are not (their means differ by block effects). The permutation distribution therefore permutes treatment labels independently within each block, giving $(k!)^b$ equally likely arrangements. Within-block shuffles leave the block means and SS_{Tot} unchanged. Hence $SS_E = SS_{\text{Tot}} - SS_{\text{Blocks}} - SS_{\text{Tr}}$ moves opposite to SS_{Tr}, and the permutation F-test is the same as the permutation test based on SS_{Tr}.

6.2.1 The Friedman test

Rank the k observations within each block. Let $R_i = \sum_j R_{ij}$ be the rank sum of treatment i and $\bar R_{i\cdot} = R_i/b$.

Definition (Friedman statistic)

$$FM = \frac{12b}{k(k+1)} \sum_{i=1}^{k} \left(\bar R_{i\cdot} - \frac{k+1}{2}\right)^2 = \frac{12}{bk(k+1)} \sum_i R_i^2 - 3b(k+1) \text{ (no ties).}$$

Under H_0 the within-block rank vectors are independent uniform permutations, so the null distribution is distribution-free, and $FM \xrightarrow{d} \chi^2_{k-1}$ as $b \to \infty$. With ties, divide by the average within-block rank variance instead of $k(k+1)/12$.

Example 6.1

$k = 3$, $b = 4$, within-block ranks of (A, B, C): $(1,2,3)$, $(1,3,2)$, $(1,2,3)$, $(2,1,3)$. The rank sums are $5, 8, 11$, so $FM = \frac{12}{48}(25 + 64 + 121) - 48 = 4.5$. The χ^2_2 approximation gives $p = e^{-2.25} = 0.105$. Enumerating all $(3!)^4 = 1296$ arrangements gives the exact $p = 162/1296 = 0.125$. With few blocks the exact distribution is noticeably different.

6.2.2 Page's trend test

For $H_1 : t_1 \le \cdots \le t_k$, use $L = \sum_{i=1}^{k} iR_i$, a sum over blocks of a Spearman-type statistic between treatment order and within-block ranks.

Proposition 6.3

Under H_0, $E(L) = \dfrac{bk(k+1)^2}{4}$ and $\mathrm{Var}(L) = \dfrac{bk^2(k+1)(k^2-1)}{144}$.

Proof. Within one block, $L_j = \sum_i iR_{ij}$ with $(R_{1j}, \ldots, R_{kj})$ a uniform permutation. Each R_{ij} has mean $\frac{k+1}{2}$, so $EL_j = \frac{k+1}{2} \cdot \frac{k(k+1)}{2}$. By Proposition 5.2 applied to $a_i = i$ and $b =$ ranks, $\mathrm{Var}\, L_j = \frac{1}{k-1}\Big(\sum_i (i - \frac{k+1}{2})^2\Big)^2 = \frac{1}{k-1}\Big(\frac{k(k^2-1)}{12}\Big)^2 = \frac{k^2(k+1)(k^2-1)}{144}$. The blocks are independent; multiply by b. ■

6.3 General two-way layouts

With c replicates per cell and no interaction, $X_{ijl} = \mu + t_i + \beta_j + \epsilon_{ijl}$. The permutation test shuffles all kc observations within each block. The rank version (**Skillings–Mack**) ranks within blocks and uses

$$SM = \frac{b\sum_i (\bar{R}_{i\cdot} - \bar{R})^2}{\text{average within-block rank variance}} \dot{\sim} \chi^2_{k-1}.$$

Again, the permutation distribution is more reliable than the χ^2 approximation in small designs. Caution: when interaction is present, a nonlinear transformation such as ranking can destroy or create interaction. Then use special aligned-rank procedures, or treat the model as a regression and use the resampling methods of Chapter 10.

Chapter 7. The Bootstrap

7.1 The plug-in principle

Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} F$ and $\hat\theta = T(X_1, \dots, X_n)$ estimate $\theta = \theta(F)$. The **sampling distribution** G_F of $\hat\theta$, and with it the bias, standard error and quantiles, depends on the unknown F. If we could draw new samples from F we would simply simulate. The bootstrap replaces F by an estimate $\hat F$ and simulates from $\hat F$:

$$\text{real world: } F \to \mathbb{X}_n \to \hat\theta, \qquad \text{bootstrap world: } \hat F \to \mathbb{X}_n^* \to \hat\theta^*.$$

The distribution of $\hat\theta^*$ given the data, $G^* = G_{\hat F}$, is the **bootstrap distribution**; it estimates G_F.

Nonparametric bootstrap. $\hat F = F_n$, the empirical distribution (mass $1/n$ at each observation). Sampling from F_n means drawing n observations *with replacement* from the data.

1. Draw a bootstrap sample $X_1^*, \dots, X_n^*$ with replacement from $\{X_1, \dots, X_n\}$ and compute $\hat\theta^* = T(X_1^*, \dots, X_n^*)$.
2. Repeat B times to get $\hat\theta^{*1}, \dots, \hat\theta^{*B}$; their empirical distribution approximates G^*.

There are two errors: $F_n \neq F$ (statistical, of order $n^{-1/2}$, the important one) and $B < \infty$ (Monte Carlo, of order $B^{-1/2}$, reducible at will).

7.2 Bootstrap estimates of standard error and bias

Replacing F by F_n in $\text{se}_F(\hat\theta)$ and $\text{bias}_F(\hat\theta) = E_F\hat\theta - \theta(F)$ gives

$$\widehat{\text{se}}_{\text{Boot}} = \Big(\frac{1}{B-1}\sum_b (\hat\theta^{*b} - \bar\theta^*)^2\Big)^{1/2}, \qquad \widehat{\text{bias}}_{\text{Boot}} = \bar\theta^* - \hat\theta, \qquad \bar\theta^* = \frac{1}{B}\sum_b \hat\theta^{*b}.$$

Note that the role of the true $\theta(F)$ is played by $\theta(F_n)$, which is $\hat\theta$ for plug-in statistics.

Example 7.1 (An exact bootstrap distribution)

Data $\{1, 4, 9\}$, $\hat\theta$ = median = 4. Among the $3^3 = 27$ equally likely resamples, the median equals the smallest value 1 iff at least two draws are 1: probability $(3 \cdot 2 + 1)/27 = 7/27$. Likewise it equals 9 with probability 7/27, and 4 with probability 13/27. Hence

$$E^*\hat\theta^* = \frac{7 \cdot 1 + 13 \cdot 4 + 7 \cdot 9}{27} = 4.52, \qquad \text{se}^* = \Big(\frac{7 + 13 \cdot 16 + 7 \cdot 81}{27} - 4.52^2\Big)^{1/2} = 2.92,$$

and the bootstrap bias estimate is 0.52. With B random resamples these values are only approximated. Here they can be computed exactly because n is tiny.

When the bootstrap works

For smooth statistics (means, smooth functions of means, many M- and U-statistics) the bootstrap distribution of $\sqrt{n}(\hat\theta^* - \hat\theta)$ consistently estimates that of $\sqrt{n}(\hat\theta - \theta)$. It can fail for non-smooth targets at the boundary of the parameter space, e.g. $\hat\theta = \max_i X_i$ for Uniform$(0, \theta)$, where $P^*(\hat\theta^* = \hat\theta) = 1 - (1 - 1/n)^n \to 1 - e^{-1}$, whereas $P(\hat\theta = \theta) = 0$.

7.3 Parametric bootstrap

If $F = F_\eta$ belongs to a parametric family, estimate $\hat{\eta}$ (e.g. by maximum likelihood) and draw bootstrap samples from $F_{\hat{\eta}}$ instead of F_n. When the model is right this is more accurate than the nonparametric bootstrap, and it is useful when the sampling distribution of $\hat{\theta}$ is analytically intractable (e.g. a nonlinear functional of a Weibull or gamma fit). When the model is wrong it can be badly biased.

7.4 Resampling designs

The guiding rule is: *resample the way the data were generated.*

- **Several samples.** For independent samples from $F_1, \ldots, F_k$, resample within each sample separately, keeping the sample sizes $n_1, \ldots, n_k$.
- **Bivariate data, random design.** If $(X_i, Y_i) \overset{\text{i.i.d.}}{\sim} F$, resample the pairs (**case** or pairs bootstrap). This keeps the dependence between X and Y and is valid even if a regression model is wrong.
- **Regression, fixed design.** If $Y_i = \alpha + \beta x_i + \epsilon_i$ with fixed x_i and i.i.d. errors, resample residuals: $Y_i^* = \hat{\alpha} + \hat{\beta} x_i + r_i^*$, with r_i^* drawn from the (leverage-adjusted) residuals. This is more efficient when the model is right but relies on it, including homoscedasticity.

Choosing between the two regression schemes:

- Classical regression inference is conditional on the x's. The residual bootstrap keeps the x's fixed, so it reproduces this setting and agrees closely with normal theory when that theory is adequate.
- The case bootstrap remains valid when the linear model is wrong. Then β is reinterpreted as the best linear approximation, the minimizer of $E(Y - \alpha - \beta X)^2$, and heteroscedasticity is allowed. The cost is some efficiency when the model is right.
- In designed experiments the x's are chosen by the experimenter, which argues for keeping them fixed.

Contrast with permutation tests: the bootstrap samples *with* replacement and keeps the structure of the alternative (e.g. the pairs (X_i, Y_i) stay together). A permutation test samples *without* replacement and deliberately destroys the association, to produce the null distribution.

Chapter 8. Bootstrap Confidence Intervals

Setting

From bootstrap replicates $\hat{\theta}^{*1}, \dots, \hat{\theta}^{*B}$ (distribution G^*, quantiles Q_{G^*}) we build a $1-\alpha$ interval for θ. The methods differ in what they assume is (approximately) a pivot.

8.1 Normal and basic intervals

Normal-approximation bootstrap interval. If $(\hat{\theta} - \text{bias} - \theta)/\,\text{se} \dot{\sim} N(0,1)$,

$$\theta \in \hat{\theta} - \widehat{\text{bias}}_{\text{Boot}} \pm z_{\alpha/2} \widehat{\text{se}}_{\text{Boot}}.$$

It uses only two summaries of G^* and is only as good as the normal approximation.

Basic interval. Assume $\hat{\theta} - \theta \sim H$ with H free of θ (a pivot) and approximate H by the law H^* of $\hat{\theta}^* - \hat{\theta}$, whose quantiles are $Q_{G^*}(p) - \hat{\theta}$. Since $P(Q_H(\frac{\alpha}{2}) \le \hat{\theta} - \theta \le Q_H(1-\frac{\alpha}{2})) = 1-\alpha$,

$$\theta \in (2\hat{\theta} - Q_{G^*}(1-\tfrac{\alpha}{2}),\ 2\hat{\theta} - Q_{G^*}(\tfrac{\alpha}{2})). \tag{8.1}$$

8.2 The bootstrap-t interval

Assume instead that the studentized quantity $t = (\hat{\theta} - \theta)/\widehat{\text{se}}$ is (approximately) pivotal, and approximate its law by that of

$$t^* = \frac{\hat{\theta}^* - \hat{\theta}}{\widehat{\text{se}}^*},$$

where $\widehat{\text{se}}^*$ is the standard error estimate recomputed on each bootstrap sample. With Q_{t^*} the quantiles of the t^{*b},

$$\theta \in (\hat{\theta} - \widehat{\text{se}}\, Q_{t^*}(1-\tfrac{\alpha}{2}),\ \hat{\theta} - \widehat{\text{se}}\, Q_{t^*}(\tfrac{\alpha}{2})). \tag{8.2}$$

If $\widehat{\text{se}}$ has a formula (e.g. $s/\sqrt{n}$ for a mean, or regression standard errors), this is cheap. Otherwise each $\widehat{\text{se}}^*$ needs a second-level bootstrap ($B \times C$ resamples, with $C \approx 25$ usually enough). The method works best on a scale where t is close to pivotal. E.g. for a variance, apply it to $\log \hat{\theta}$ and exponentiate the endpoints; an increasing transformation preserves coverage.

8.3 Percentile intervals

Motivation. The bootstrap distribution of an estimator approximates its sampling distribution, so its quantiles give a confidence interval without any formula for the standard error. This section studies when the simple percentile interval works.

Percentile interval. $\theta \in (Q_{G^*}(\frac{\alpha}{2}), Q_{G^*}(1-\frac{\alpha}{2}))$.

Proposition 8.1

Suppose there is an increasing φ, not necessarily known, with $\varphi(\hat{\theta}) - \varphi(\theta) \sim N(0,1)$, and the same holds in the bootstrap world: $\varphi(\hat{\theta}^*) - \varphi(\hat{\theta}) \sim N(0,1)$ given the data. Then the

percentile interval is exact.

Proof. The exact interval is $\varphi(\theta) \in \varphi(\hat\theta) \pm z_{\alpha/2}$, i.e. $\theta \in (\varphi^{-1}(\varphi(\hat\theta) - z_{\alpha/2}), \varphi^{-1}(\varphi(\hat\theta) + z_{\alpha/2}))$. In the bootstrap world $\varphi(\hat\theta^*) \sim N(\varphi(\hat\theta), 1)$, so $\varphi(\hat\theta) \pm z_{\alpha/2}$ are its $\frac{\alpha}{2}$ and $1 - \frac{\alpha}{2}$ quantiles. Since φ is increasing, applying φ^{-1} maps them to the quantiles $Q_{G^*}(\frac{\alpha}{2})$, $Q_{G^*}(1 - \frac{\alpha}{2})$ of $\hat\theta^*$. The unknown φ never has to be computed. ■

The percentile interval is transformation-respecting and always lies in the parameter space. Its weak point is the assumption that $\varphi(\hat\theta)$ is median-unbiased with constant variance.

8.3.1 Bias-corrected and accelerated (BCa) intervals

Generalize the assumption to

$$\varphi(\hat\theta) - \varphi(\theta) \sim N(-b\,\sigma_\varphi,\ \sigma_\varphi^2), \qquad \sigma_\varphi = 1 + a\,\varphi(\theta), \tag{8.3}$$

with a bias constant b and an acceleration a (the rate at which the standard error changes with the parameter).

Proposition 8.2

Under (8.3) and its bootstrap analogue, the interval $(Q_{G^*}(p_l), Q_{G^*}(p_u))$ is exact, where with $z_l = \Phi^{-1}(\frac{\alpha}{2})$, $z_u = \Phi^{-1}(1 - \frac{\alpha}{2})$,

$$p_l = \Phi\Big(b + \frac{b + z_l}{1 - a(b + z_l)}\Big), \qquad p_u = \Phi\Big(b + \frac{b + z_u}{1 - a(b + z_u)}\Big).$$

Proof [Supplement]. Write $\hat\varphi = \varphi(\hat\theta)$, $\phi = \varphi(\theta)$, and assume $1 + a\phi > 0$. Then $Z = \frac{\hat\varphi - \phi}{1 + a\phi} + b \sim N(0, 1)$. Solving, $\phi = g(Z) = \frac{\hat\varphi - (Z - b)}{1 + a(Z - b)}$ with $g'(Z) = -\frac{1 + a\hat\varphi}{(1 + a(Z-b))^2} < 0$. So $\phi \in [g(-z_l), g(-z_u)] = [L, U]$ iff $Z \in [z_l, z_u]$, which has probability $1 - \alpha$; here $L = \frac{\hat\varphi + b + z_l}{1 - a(b + z_l)}$ and similarly U. In the bootstrap world, $\varphi(\hat\theta^*) \sim N(\mu_*, \sigma_*^2)$ with $\mu_* = \hat\varphi - b(1 + a\hat\varphi)$, $\sigma_* = 1 + a\hat\varphi$. With $w = b + z_l$, $L - \hat\varphi = \frac{w(1 + a\hat\varphi)}{1 - aw}$, so $P^*(\varphi(\hat\theta^*) \le L) = \Phi(\frac{L - \mu_*}{\sigma_*}) = \Phi(\frac{w}{1 - aw} + b) = p_l$. Thus $\varphi^{-1}(L) = Q_{G^*}(p_l)$, and similarly for U. ■

Estimating b and a. Under the model, $P^*(\hat\theta^* \le \hat\theta) = \Phi(b)$, so

$$\hat b = \Phi^{-1}\Big(\frac{\#\{\hat\theta^{*b} \le \hat\theta\}}{B}\Big).$$

For a, use the jackknife: with $\hat\theta_{(i)}$ computed without observation i and $\hat\theta_{(\cdot)}$ their mean,

$$\hat a = \frac{\sum_i (\hat\theta_{(\cdot)} - \hat\theta_{(i)})^3}{6\big[\sum_i (\hat\theta_{(\cdot)} - \hat\theta_{(i)})^2\big]^{3/2}},$$

a skewness measure of the influence of each observation. The jackknife (leave-one-out recomputation; Quenouille 1949, Tukey 1958) was originally proposed for bias and standard-error estimation and is a precursor of the bootstrap. With $a = 0$ the method reduces to the **bias-corrected (BC)** interval, with $p = \Phi(2\hat b + z)$.

8.4 Accuracy and recommendations

For a smooth statistic, the coverage of an interval C satisfies $P(\theta \in C) = 1 - \alpha + O(n^{-k/2})$:

- first-order accurate ($k = 1$): normal theory, normal bootstrap, basic, percentile, BC;
- second-order accurate ($k = 2$): bootstrap-t, BCa.

Recommendation: bootstrap-t when a reliable standard error formula exists (it then needs no second-level resampling); otherwise BCa. Use $B \geq 2000$ for intervals, since tail quantiles are needed. The percentile interval is the simplest default and is often adequate for large samples. For multi-sample or regression resampling, the jackknife estimate of a is less reliable.

Chapter 9. Bootstrap Tests

The problem

A test needs the distribution of the statistic *under* H_0, but the bootstrap mimics the *actual* F, which may not satisfy H_0. There are two ways around this.

1. **Resample under the null.** Build an estimate $\hat{F}_0$ that satisfies H_0, and resample from it.
2. **Resample a pivot.** If $T_\theta = (\hat{\theta} - \theta)/\widehat{\text{se}}$ is (approximately) pivotal, its law is the same under H_0 and under the truth. Approximate it by $T^* = (\hat{\theta}^* - \hat{\theta})/\widehat{\text{se}}^*$ resampled from $\hat{F}$, and compare with the observed T_{θ_0}.

In both cases the p-value is $\hat{p} = \{1 + \#(T^{*b} \text{ at least as extreme as } T_{\text{obs}})\}/(B+1)$.

9.1 Parametric bootstrap tests

In a model $F(\cdot; \theta, \eta)$ with nuisance η, test $H_0 : \theta = \theta_0$ with a statistic $S(\mathbb{X}, \theta_0)$ that is large under H_1.

- *Null resampling:* compute the restricted maximum likelihood estimate $\hat{\eta}_0$ under $\theta = \theta_0$, draw samples from $F(\cdot; \theta_0, \hat{\eta}_0)$, and recompute $S(\mathbb{X}^*, \theta_0)$.
- *Pivot resampling:* draw from $F(\cdot; \hat{\theta}, \hat{\eta})$ and recompute $S(\mathbb{X}^*, \hat{\theta})$.

Proposition 9.1

For $X_i \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$ and $S = \sqrt{n}(\bar{X} - \mu_0)/s$, both schemes reproduce the exact null distribution t_{n-1}.

Proof. Under null resampling the bootstrap sample is $N(\mu_0, \hat{\sigma}_0^2)$, so $\sqrt{n}(\bar{X}^* - \mu_0)/s^* \sim t_{n-1}$ whatever $\hat{\sigma}_0$ is. Under pivot resampling the sample is $N(\hat{\mu}, \hat{\sigma}^2)$ and $\sqrt{n}(\bar{X}^* - \hat{\mu})/s^* \sim t_{n-1}$. ■

Exact pivots are not always available. For Laplace(μ, b), $|X_i - \mu|/b \sim \text{Exp}(1)$, so $2\sum_i |X_i - \mu|/b \sim \chi^2_{2n}$ is a pivot for μ only if b is known. With b unknown one falls back on approximate pivots (e.g. the CLT-based t-statistic), or on null resampling from Laplace$(\mu_0, \hat{b}_0)$ with $\hat{b}_0 = \frac{1}{n}\sum |X_i - \mu_0|$. With two samples and a common scale, the null model fits a common center (the pooled median) and a pooled $\hat{b}_0$.

In general the two schemes agree asymptotically. Studentized statistics give more accurate tests than raw differences (Hall and Wilson). A non-pivotal statistic, such as a ratio $\hat{\theta}/\theta_0$ whose spread depends on θ, should only be used with null resampling.

9.2 Nonparametric bootstrap tests

One sample, location. For $H_0 : \theta = \theta_0$ in $X_i = \theta + \epsilon_i$:

- null resampling: resample from the recentered data $\tilde{X}_i = X_i - \hat{\theta} + \theta_0$, which satisfy H_0;
- pivot resampling: resample from the raw data and use $T^* = (\hat{\theta}^* - \hat{\theta})/\widehat{\text{se}}^*$.

For a mean and the t-statistic the two are identical: recentering shifts every bootstrap mean by the same constant. A two-sided test uses $|T^*| \geq |T_{\text{obs}}|$.

Two samples, $H_0 : F = G$. Pool the samples and draw both bootstrap samples from the pooled empirical distribution. This is the with-replacement analogue of the permutation test, and the two give similar answers except in very small samples.

Two samples, equal locations with possibly different shapes. Here $F \neq G$ is allowed under $H_0 : \mu_1 = \mu_2$, so pooling is wrong and permutation tests are invalid. Use pivot resampling within each sample:

$$T^* = \frac{\bar{Y}^* - \bar{X}^* - (\bar{Y} - \bar{X})}{\sqrt{s_X^{*2}/m + s_Y^{*2}/n}}, \qquad T_{\text{obs}} = \frac{\bar{Y} - \bar{X}}{\sqrt{s_X^2/m + s_Y^2/n}}.$$

Equivalently, recenter each sample to the common pooled mean and resample within samples. Centering both samples at a common value and then using the *unstandardized* difference without subtracting $\bar{Y} - \bar{X}$ does not give a null distribution; it reproduces the sampling distribution of the observed difference.

9.3 Duality with bootstrap intervals

Motivation. As in classical inference, a test can be obtained by checking whether the null value falls inside a confidence interval, so every bootstrap interval yields a bootstrap test.

Proposition 9.2

The two-sided level-α pivot test of $H_0 : \theta = \theta_0$ based on $\hat{\theta} - \theta$ (resp. on $(\hat{\theta} - \theta)/\widehat{\text{se}}$) accepts exactly when θ_0 lies in the basic (resp. bootstrap-t) interval.

Proof. The test accepts iff $Q_{H^*}(\frac{\alpha}{2}) \leq \hat{\theta} - \theta_0 \leq Q_{H^*}(1 - \frac{\alpha}{2})$, i.e. iff $\hat{\theta} - Q_{H^*}(1 - \frac{\alpha}{2}) \leq \theta_0 \leq \hat{\theta} - Q_{H^*}(\frac{\alpha}{2})$, which is (8.1). The studentized case gives (8.2) in the same way. ■

Percentile-type intervals have no simple test counterpart, because the transformation φ behind them is unknown.

Chapter 10. Linear Regression

10.1 Simple linear regression

Motivation. Least squares is sensitive to outliers in y. Rank-based slope estimators, such as the median of pairwise slopes, give robust alternatives with distribution-free confidence intervals.

Model $y_i = \alpha + \beta x_i + \epsilon_i$ with uncorrelated errors of mean 0 and variance σ^2; inference is conditional on the x_i. Least squares gives $\hat\beta = S_{xy}/S_{xx}$, $\hat\alpha = \bar y - \hat\beta\bar x$ and $s^2 = \sum e_i^2/(n-2)$.

Proposition 10.1

$E\hat\beta = \beta$ and $\operatorname{Var}\hat\beta = \sigma^2/S_{xx}$. The residuals satisfy $\operatorname{Var}(e_i) = \sigma^2(1-h_i)$ with leverage $h_i = \frac{1}{n} + \frac{(x_i-\bar x)^2}{S_{xx}}$, so the adjusted residuals $r_i = e_i/\sqrt{1-h_i}$ have variance σ^2.

Proof. $\hat\beta = \sum w_i y_i$ with $w_i = (x_i - \bar x)/S_{xx}$, $\sum w_i = 0$, $\sum w_i x_i = 1$, $\sum w_i^2 = 1/S_{xx}$. With the hat matrix H, $\mathbf{e} = (I-H)\boldsymbol\epsilon$ and $\operatorname{Var}\mathbf{e} = \sigma^2(I-H)$, whose diagonal entries are $\sigma^2(1-h_i)$. ■

Under normal errors, $(\hat\beta - \beta)/\widehat{\mathrm{se}}(\hat\beta) \sim t_{n-2}$ with $\widehat{\mathrm{se}}(\hat\beta) = s/\sqrt{S_{xx}}$. Without normality we use resampling.

Permutation test of $H_0 : \beta = 0$. Under H_0, $y_i = \alpha + \epsilon_i$ are exchangeable, so shuffle the y's against the fixed x's. The statistics $\hat\beta$, $\sum x_i y_i$ and $t = \hat\beta/\widehat{\mathrm{se}}(\hat\beta)$ give the same test: with S_{xx} and S_{yy} fixed, $t = \hat\beta\sqrt{(n-2)S_{xx}}/\sqrt{S_{yy} - \hat\beta^2 S_{xx}}$ is increasing in $\hat\beta$. This is the same test as the permutation test of correlation (Section 5.1).

Bootstrap intervals. Use the case bootstrap (random x, robust to model failure) or the residual bootstrap $y_i^* = \hat\alpha + \hat\beta x_i + r_i^*$ (fixed design, model assumed). Because $\widehat{\mathrm{se}}(\hat\beta)$ has a formula, the bootstrap-t interval

$$\beta \in \big(\hat\beta - \widehat{\mathrm{se}}(\hat\beta)Q_{t^*}(1-\tfrac{\alpha}{2}),\ \hat\beta - \widehat{\mathrm{se}}(\hat\beta)Q_{t^*}(\tfrac{\alpha}{2})\big), \qquad t^* = \frac{\hat\beta^* - \hat\beta}{\widehat{\mathrm{se}}(\hat\beta^*)},$$

is cheap and second-order accurate. For tests of $\beta = 0$:

- null resampling: resample x and y *independently* (case form), or $y^* = \bar y + r_0^*$ with $r_{0,i} = (y_i - \bar y)/\sqrt{1-1/n}$ (residual form);
- pivot resampling: use t^* above and compare with $t_{\mathrm{obs}} = \hat\beta/\widehat{\mathrm{se}}(\hat\beta)$, which inverts the bootstrap-t interval.

10.2 Multiple regression

$\mathbf{y} = X\boldsymbol\beta + \boldsymbol\epsilon$ with X of size $n \times (p+1)$: $\hat{\boldsymbol\beta} = (X'X)^{-1}X'\mathbf{y}$, $E\hat{\boldsymbol\beta} = \boldsymbol\beta$, $\operatorname{Var}\hat{\boldsymbol\beta} = \sigma^2(X'X)^{-1}$, $s^2 = \mathbf{e}'\mathbf{e}/(n-p-1)$ (unbiased since $E\,\boldsymbol\epsilon'(I-H)\boldsymbol\epsilon = \sigma^2\operatorname{tr}(I-H)$). Each β_j is a *partial* effect: the effect of x_j adjusted for the other regressors.

Comparing nested models. Split $X = [X_0, X_1]$ ($k+1$ and $q = p-k$ columns) and test $H_0 : \boldsymbol{\beta}_1 = 0$. With hat matrices H and H_0,

$$F = \frac{\mathbf{y}'(H - H_0)\mathbf{y}/q}{\mathbf{y}'(I - H)\mathbf{y}/(n-p-1)} = \frac{\hat{\boldsymbol{\beta}}_1'\{X_1'(I-H_0)X_1\}\hat{\boldsymbol{\beta}}_1}{q\,s^2}. \tag{10.1}$$

Proposition 10.2

Under H_0 with normal errors, $F \sim F_{q,n-p-1}$. More generally $F_{\boldsymbol{\beta}_1}$, defined by replacing $\hat{\boldsymbol{\beta}}_1$ with $\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta}_1$ in (10.1), has this distribution for every $\boldsymbol{\beta}_1$, so it is a pivot.

Proof. $H - H_0$ is a projection of rank q orthogonal to $I - H$, and both annihilate X_0. Under H_0, the numerator and denominator quadratic forms reduce to $\boldsymbol{\epsilon}'(H-H_0)\boldsymbol{\epsilon}$ and $\boldsymbol{\epsilon}'(I-H)\boldsymbol{\epsilon}$, independent $\sigma^2\chi^2_q$ and $\sigma^2\chi^2_{n-p-1}$ (Cochran). The second form of (10.1) is Frisch–Waugh–Lovell: $\hat{\boldsymbol{\beta}}_1$ is the regression of $(I-H_0)\mathbf{y}$ on $\tilde{X}_1 = (I-H_0)X_1$, and $H - H_0$ projects onto the column space of $\tilde{X}_1$. For general $\boldsymbol{\beta}_1$, apply the same argument to $\mathbf{y} - X_1\boldsymbol{\beta}_1$. ■

For $q = 1$, $F = t_j^2$, so the F-test of one coefficient is the two-sided t-test. A one-sided alternative $\beta_j > 0$ needs t_j itself, and the permutation and bootstrap schemes below apply to t_j in the same way. For bootstrap tests in multiple regression, null resampling uses either the case form (draw the $\mathbf{x}$-rows and the y's independently, which is appropriate when all slopes are tested) or residuals of the reduced model with replacement. Pivot resampling uses the case or full-model residual bootstrap with $F_{\hat{\boldsymbol{\beta}}_1}$ or $t_j^* = (\hat{\beta}_j^* - \hat{\beta}_j)/\widehat{\mathrm{se}}(\hat{\beta}_j^*)$, and the p-value is $\#\{F^{*b} \geq F_{\mathrm{obs}}\}/(B+1)$ (with the observed value counted).

Permutation tests for $H_0 : \boldsymbol{\beta}_1 = 0$. Under H_0 the errors are exchangeable but the y_i are not (their means depend on X_0), so no permutation test is exact. Three approximate schemes:

1. *Shuffle y:* effectively tests $\beta_1 = \cdots = \beta_p = 0$; it can be anticonservative and is not recommended.
2. *Freedman–Lane:* fit the reduced model, shuffle its residuals e_0, set $y^* = \hat{y}_0 + e_0^*$, and recompute F. This is generally the most accurate.
3. *ter Braak:* shuffle the full-model residuals, set $y^* = \hat{y} + e^*$, and recompute the pivot $F_{\hat{\boldsymbol{\beta}}_1}$.

All three are asymptotically equivalent. Replacing "shuffle" with "resample with replacement" in schemes 2 and 3 gives the residual bootstrap tests under the null and with the pivot, respectively. Bootstrap-t intervals for each β_j use $t_j^* = (\hat{\beta}_j^* - \hat{\beta}_j)/\widehat{\mathrm{se}}(\hat{\beta}_j^*)$ with $\widehat{\mathrm{se}}^2(\hat{\beta}_j) = s^2(X'X)^{-1}_{jj}$.

10.3 Robust regression

Least squares is the maximum likelihood estimator under normal errors. It loses efficiency under heavy-tailed errors and can be ruined by a single outlier.

Definition (Breakdown point)

The smallest fraction of observations that, sent to infinity, can send the estimator to infinity. Roughly: the mean has $1/n \to 0$, the α-trimmed mean α, and the median $1/2$.

Trimmed and Winsorized means. The α-trimmed mean drops the lowest and highest α fractions of the data and averages the rest. The α-Winsorized mean instead replaces them by the α and $1-\alpha$ sample quantiles and then averages all n values. Both have breakdown point α. The Hodges–Lehmann estimator has breakdown point $1-1/\sqrt{2} \approx 29\%$: sending a fraction ε of points to infinity drags a fraction $1-(1-\varepsilon)^2$ of the Walsh averages with them, which exceeds $\frac{1}{2}$ when $\varepsilon > 1-1/\sqrt{2}$.

Example 10.1

For $-2,-1,-1,0,0,1,1,2,3,50$: mean 5.3; 10% trimmed mean 0.625 (drop -2 and 50); 10% Winsorized mean 0.7 (replace them by -1 and 3); median 0.5. Replacing 50 by 10^{100} changes only the mean.

Trimming and Winsorizing treat both tails alike, even when the contamination is one-sided. M-estimators instead down-weight observations according to their own residuals.

10.3.1 M-estimators

For location, the least squares estimator $\bar{x} = \arg\min_\mu \sum (x_i-\mu)^2$ is generalized to

$$\hat{\mu}^\Psi = \arg\min_\mu \sum_i \Psi\Big(\frac{x_i-\mu}{\hat{\tau}}\Big) \iff \sum_i \psi\Big(\frac{x_i-\mu}{\hat{\tau}}\Big) = 0, \qquad \psi = \Psi',$$

with a robust scale $\hat{\tau} = 1.4826$ MAD (consistent for σ under normality, since $\Phi^{-1}(3/4) = 0.6745$).

- $\psi(x) = x$: the mean. $\psi(x) = \operatorname{sgn}(x)$: the median.
- **Huber:** $\psi(x) = \max(-k, \min(k, x))$, quadratic Ψ in the center and linear in the tails. With $k \approx 1.35$ it has about 95% efficiency at the normal.
- **Tukey biweight:** $\psi(x) = x(1-(x/c)^2)^2 I(|x| \le c)$, redescending, so gross outliers get zero weight.

A bounded ψ with a 50% breakdown scale gives breakdown point 50%. ("M" stands for maximum-likelihood type: with $\Psi(x) = x^2/2$ and $\hat{\tau} = \sigma$, $\hat{\mu}^\Psi$ is the normal maximum likelihood estimator.) The equation has no closed form and is solved numerically, e.g. by iteratively reweighted least squares with weights $\psi(u_i)/u_i$.

Inference on a location with outliers. With a contaminated sample, the t-test and t-interval are driven by the outliers: the mean and s both inflate. Robust alternatives:

- the sign test and order-statistic interval (Chapter 2), valid for the median under any continuous F;
- the signed-rank test and Walsh-average interval, valid if the uncontaminated distribution is symmetric;
- an M-estimator with a bootstrap standard error and bias, giving a normal-approximation bootstrap interval, or better a percentile or BCa interval. Its asymptotic standard error requires the influence function, so the bootstrap is the practical route.

The lower the breakdown point of the underlying estimator, the wider the resulting interval tends to be under contamination. In regression, $\hat{\boldsymbol{\beta}}^\Psi = \arg\min \sum \Psi((y_i - \mathbf{x}_i'\boldsymbol{\beta})/\hat{\tau})$. The estimating equations $\sum \psi(\cdot)\,\mathbf{x}_i = 0$ show that Huber's estimator is robust to outlying y but not to outlying x (high leverage), because the factor x_{ij} is unbounded. **Yohai's MM-estimator** uses a redescending ψ

(biweight, $c = 4.685$ for 95% efficiency) started from a high-breakdown S-estimate of scale, achieving a 50% breakdown point together with high efficiency. **Least trimmed squares** minimizes the sum of the h smallest squared residuals; $h = \lfloor n/2 \rfloor + \lfloor (p+1)/2 \rfloor$ gives maximal breakdown, and $h = \lfloor 0.9n \rfloor$ a 10% trimming. **Least median of squares** minimizes the median squared residual. Both are very robust but inefficient, and are mainly used as starting values.

10.3.2 The Theil–Sen estimator

For $y_i = \alpha + \beta x_i + \epsilon_i$, form the pairwise slopes $\beta_{ij} = (y_j - y_i)/(x_j - x_i)$ for $x_i \neq x_j$ and set

$$\hat{\beta}^{TS} = \text{median}_{i<j}\, \beta_{ij}, \qquad \hat{\alpha}^{TS} = \text{median}_i (y_i - \hat{\beta}^{TS} x_i).$$

Proposition 10.3 (Link to Kendall's tau)

Let $\tau_n(\beta_0)$ be Kendall's statistic between x_i and $y_i - \beta_0 x_i$. Then $\binom{n}{2}\tau_n(\beta_0) = \#\{\beta_{ij} > \beta_0\} - \#\{\beta_{ij} < \beta_0\}$. Hence the test of $H_0 : \beta = \beta_0$ by Kendall's tau, and the interval $(\beta_{(l)}, \beta_{(u)})$ with $l = \binom{n}{2} \frac{Q_K(\alpha/2)+1}{2}$, $u = \binom{n}{2} \frac{Q_K(1-\alpha/2)+1}{2}$ (Q_K the null quantile of τ_n), are exact. $\hat{\beta}^{TS}$ is the corresponding point estimate.

Proof. $\text{sgn}[(x_i - x_j)\{(y_i - \beta_0 x_i) - (y_j - \beta_0 x_j)\}] = \text{sgn}[(x_i - x_j)^2(\beta_{ij} - \beta_0)] = \text{sgn}(\beta_{ij} - \beta_0)$. At the true β, $y_i - \beta x_i = \alpha + \epsilon_i$ is independent of x_i, so $\tau_n(\beta)$ has the null law K_n. The acceptance region $Q_K(\frac{\alpha}{2}) \le \tau_n(\beta_0) \le Q_K(1 - \frac{\alpha}{2})$ is a range of $\#\{\beta_{ij} > \beta_0\}$, which is a step function of β_0 dropping by one at each ordered slope. So the region is an interval of ordered slopes with the stated indices. ■

$\hat{\beta}^{TS}$ has breakdown point $1 - 1/\sqrt{2} \approx 29\%$ and about 96% ARE relative to least squares under normal errors.

Example 10.2

Points $(1,1), (2,2), (3,3.5), (4,4), (5,20)$, where the last point is an outlier. The ten ordered pairwise slopes are $0.5, 1, 1, 1, 1.25, 1.5, 4.75, 6, 8.25, 16$, so $\hat{\beta}^{TS} = 1.375$ and $\hat{\alpha}^{TS} = -0.625$. Least squares gives slope 4.0, pulled up by a single point.

In multiple regression Theil–Sen has no unique definition. One version uses **backfitting**:

1. Start with single-variable Theil–Sen slopes $\hat{\beta}_j^{(0)}$ and set $\hat{\beta}_0^{(0)} = \text{median}_i(y_i - \sum_j \hat{\beta}_j^{(0)} x_{ij})$.
2. For $k = 1, 2, \ldots$ and each j, compute the Theil–Sen slope of the partial responses $y_i - \sum_{l \neq j} \hat{\beta}_l^{(k-1)} x_{il}$ on x_{ij}, then update the intercept as a median.
3. Iterate until the coefficients stabilize.

Inference with robust regression estimators. M- and MM-estimates come with approximate standard errors from the final weighted least squares step. These are reasonable for large n, and the bootstrap standard errors are usually somewhat larger and more reliable. With an approximate standard error one can form a bootstrap-t interval $(\hat{\beta}_j - \widehat{\text{se}}\, Q_{t^*}(1 - \frac{\alpha}{2}), \hat{\beta}_j - \widehat{\text{se}}\, Q_{t^*}(\frac{\alpha}{2}))$ from case resampling, or permutation tests of $t_j = \hat{\beta}_j / \widehat{\text{se}}(\hat{\beta}_j)$. Shuffling y alone tests the joint null that all slopes vanish; for a single coefficient use a residual-shuffling scheme as in Section 10.2. Bootstrap samples contain many repeated points, which can upset estimators based on pairwise slopes (ties in x); permutation tests avoid this.

Chapter 11. Smoothing and Density Estimation

11.1 From histograms to kernels

The empirical distribution function estimates F well but has no derivative, so it gives no direct estimate of the density f. A histogram does, but it depends on both the bin width and the bin origin. Averaging histograms over shifted origins gives a smooth estimate: the kernel estimator.

Definition (Kernel density estimator)

For a kernel K and bandwidth $h > 0$, $\hat{f}(x) = \frac{1}{nh}\sum_{i=1}^{n} K\Big(\frac{x - X_i}{h}\Big)$.

Standard kernels satisfy $K \geq 0$, $\int K = 1$, $K(-u) = K(u)$, and $\sigma_K^2 = \int u^2 K < \infty$. Examples are the Gaussian ϕ, the Epanechnikov kernel $\frac{3}{4}(1-u^2)_+$ and the uniform kernel $\frac{1}{2}I(|u| \leq 1)$. Then $\hat{f} \geq 0$ and $\int \hat{f} = 1$ (substitute $u = (x - X_i)/h$), so $\hat{f}$ is a density. The choice of h matters far more than the choice of K.

Example 11.1

Data $\{0, 1, 3\}$, Gaussian kernel, $h = 1$: $\hat{f}(1) = \frac{1}{3}[\phi(1)+\phi(0)+\phi(2)] = \frac{1}{3}(0.242+0.399+0.054) = 0.232$. The point 3, two bandwidths away, contributes little.

11.2 Bias, variance and bandwidth

Motivation. A kernel density estimate is controlled by its bandwidth: small bandwidths give noisy estimates, large ones oversmooth. Computing bias and variance shows how the optimal bandwidth shrinks with n and gives the $n^{-4/5}$ rate.

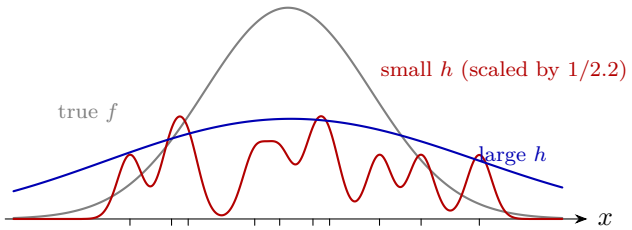


Figure. *Kernel density estimates from the same ten points (ticks). A small bandwidth gives a spiky estimate with large variance (its vertical scale is shrunk here to fit); a large bandwidth flattens the peak, a bias. The optimal $h \propto n^{-1/5}$ balances the two.*

Theorem 11.1

Let f be twice continuously differentiable near x with $f(x) > 0$. As $n \to \infty$, $h \to 0$ and $nh \to \infty$,

$$\operatorname{bias}\hat{f}(x) = \frac{\sigma_K^2 f''(x)}{2}h^2 + o(h^2), \qquad \operatorname{Var}\hat{f}(x) = \frac{f(x)R(K)}{nh} + o\Big(\frac{1}{nh}\Big), \qquad R(K) = \int K^2.$$

Proof [Supplement]. $E\hat{f}(x) = \int \frac{1}{h}K(\frac{x-t}{h})f(t)\,dt = \int K(u)f(x-hu)\,du$. Expand $f(x-hu) = f(x) - huf'(x) + \frac{h^2u^2}{2}f''(x) + o(h^2u^2)$, use $\int uK = 0$ and $\int u^2K = \sigma_K^2$, and justify the remainder

by dominated convergence. For the variance, $\operatorname{Var}\hat f(x) = \frac{1}{n}\operatorname{Var}\left(\frac{1}{h}K(\frac{x-X}{h})\right)$. Here $E\frac{1}{h^2}K^2(\frac{x-X}{h}) = \frac{1}{h}\int K^2(u)f(x-hu)\,du = \frac{f(x)R(K)}{h} + O(1)$, while the squared mean is $O(1)$. ■

This is the **bias–variance trade-off**: a small h gives low bias and high variance (a wiggly estimate), and a large h the reverse. Integrating,

$$\operatorname{MISE}(\hat f) = E\int(\hat f - f)^2 \approx \frac{\sigma_K^4 R(f'')}{4}h^4 + \frac{R(K)}{nh}, \qquad R(f'') = \int (f'')^2,$$

minimized at

$$h^* = \Big(\frac{R(K)}{\sigma_K^4 R(f'')\,n}\Big)^{1/5} \asymp n^{-1/5}, \qquad \operatorname{MISE}(h^*) \asymp n^{-4/5}. \tag{11.1}$$

(Set the derivative $\sigma_K^4 R(f'')h^3 - R(K)/(nh^2)$ to zero.) The rate $n^{-4/5}$ is slower than the parametric rate n^{-1}: this is the price of not assuming a shape.

Choosing h. The optimal h^* contains the unknown $R(f'')$.

- **Normal reference.** If $f = N(\mu,\sigma^2)$ then $R(f'') = \frac{3}{8\sqrt{\pi}\sigma^5}$, so with the Gaussian kernel $h = 1.06\,\hat\sigma n^{-1/5}$, where $\hat\sigma = \min(s, \mathrm{IQR}/1.34)$.
- **Silverman's rule of thumb:** $h = 0.9\,\hat\sigma n^{-1/5}$, slightly smaller to protect against multimodality.
- **Plug-in (Sheather–Jones):** estimate $R(f'')$ by a pilot kernel estimate of f'' and substitute into (11.1). This is generally the recommended choice; cross-validation is an alternative.

Proof of the normal value of $R(f'')$. For $\sigma = 1$, $\phi''(x) = (x^2-1)\phi(x)$ and $\phi^2 = \frac{1}{2\sqrt{\pi}} \times$ the $N(0,\frac12)$ density. So $R(\phi'') = \frac{1}{2\sqrt{\pi}}E(Z^2-1)^2$ with $Z\sim N(0,\frac12)$, which is $\frac{1}{2\sqrt{\pi}}(\frac34 - 1 + 1) = \frac{3}{8\sqrt{\pi}}$. Scaling gives the factor σ^{-5}. With $R(\phi) = \frac{1}{2\sqrt{\pi}}$ and $\sigma_K = 1$, (11.1) gives $h^* = (4/3)^{1/5}\sigma n^{-1/5} = 1.06\sigma n^{-1/5}$. ■

11.3 Multivariate density estimation

For $\mathbf{X}_i \in \mathbb{R}^p$, $\hat f(\mathbf{x}) = \frac{1}{n}\sum_i K_H(\mathbf{x}-\mathbf{X}_i)$ with $K_H(\mathbf{x}) = |H|^{-1/2}K(H^{-1/2}\mathbf{x})$ and a bandwidth matrix H (e.g. $H = \operatorname{diag}(h_1^2,\dots,h_p^2)$, or a plug-in estimate of a full matrix that follows the shape of the data). With a product kernel and equal widths h,

$$\operatorname{MSE}\hat f(\mathbf{x}) \approx \Big[\frac{\sigma_K^2 h^2}{2}\sum_j \partial_{jj} f(\mathbf{x})\Big]^2 + \frac{f(\mathbf{x})R(K)^p}{nh^p},$$

so $h^* \asymp n^{-1/(p+4)}$ and $\operatorname{MSE} \asymp n^{-4/(p+4)}$. This is the **curse of dimensionality**: in $p = 10$ dimensions, a window of side 0.1 in the unit cube contains on average $10^{-10}n$ points. Kernel density estimation is practical in one or two dimensions, occasionally three.

11.4 Confidence intervals and bands

Since bias $\hat f(x)$ and se $\hat f(x)$ are of the same order at h^*, intervals are stated for the smoothed density $f_h(x) = E\hat f(x)$ rather than for $f(x)$.

- **Pointwise, plug-in:** by the CLT, $\sqrt{nh}(\hat f(x) - f_h(x)) \xrightarrow{d} N(0, f(x)R(K))$, so $f_h(x) \in \hat f(x) \pm z_{\alpha/2}\sqrt{\hat f(x)R(K)/(nh)}$.

- **Pointwise, bootstrap:** resample the data, recompute $\hat{f}^*$ on a grid with the same h, and use $\hat{f} \pm z_{\alpha/2}\widehat{\mathrm{se}}^*$, the basic interval $(2\hat{f} - Q^*_{1-\alpha/2}, 2\hat{f} - Q^*_{\alpha/2})$, or the symmetric interval $\hat{f} \pm Q_{1-\alpha}(|\hat{f}^* - \hat{f}|)$.
- **Simultaneous band:** with $\Delta^* = \max_{x \in \text{grid}} |\hat{f}^*(x) - \hat{f}(x)|$ and its $1-\alpha$ quantile q^*, the band $\hat{f}(x) \pm q^*$ covers f_h on the whole grid with asymptotic probability $1-\alpha$. It is much wider than the pointwise intervals.

Negative lower limits can be truncated at 0 without changing coverage.

The quantity $(\hat{f}(x) - f_h(x))/\sqrt{\hat{f}(x)R(K)/(nh)}$ is asymptotically pivotal, which suggests a bootstrap-t interval with $t^* = (\hat{f}^* - \hat{f})/\sqrt{\hat{f}^* R(K)/(nh)}$. In practice it is unstable: where $\hat{f}^*(x) \approx 0$ the denominator vanishes, t^* explodes, and the interval becomes absurdly wide. The same problem rules out bands built from studentized maxima. In practice, pointwise intervals on a grid are reported more often than simultaneous bands. Some authors call such a collection a "confidence band", so check which is meant.

Chapter 12. Nonparametric Regression

Model

$Y_i = m(X_i) + \sigma(X_i)\epsilon_i$ with $E\epsilon_i = 0$, $\operatorname{Var}\epsilon_i = 1$ and $m(x) = E(Y \mid X = x)$ smooth but otherwise unrestricted. The goal is to estimate the curve m and quantify its uncertainty.

12.1 Local polynomial regression

Motivation. To estimate a regression function without a parametric form, fit a low-degree polynomial in a window around each point, weighting nearby observations more. Local linear fits correct the boundary bias of local averages.

Definition (Local polynomial estimator)

With $K_h(u) = K(u/h)/h$, fit at each x

$$\hat{\boldsymbol{\beta}}_x = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} \big(Y_i - \beta_0 - \beta_1(X_i - x) - \cdots - \beta_p(X_i - x)^p\big)^2 K_h(x - X_i),$$

and set $\hat{m}_p(x) = \hat{\beta}_{0,x}$. In matrix form, $\hat{\boldsymbol{\beta}}_x = (X_x' W_x X_x)^{-1} X_x' W_x \mathbf{y}$, with $W_x = \operatorname{diag}(K_h(x - X_i))$ and X_x having rows $(1, X_i - x, \ldots, (X_i - x)^p)$.

Every local polynomial estimator is linear in the responses: $\hat{m}_p(x) = \sum_i w_i^p(x) Y_i$.

Proposition 12.1

With $S_q = \sum_j K_h(x - X_j)(X_j - x)^q$:

(i) $p = 0$ (**Nadaraya–Watson**): $w_i^0(x) = K_h(x - X_i) / \sum_j K_h(x - X_j)$;

(ii) $p = 1$ (**local linear**): $w_i^1(x) = \dfrac{\{S_2 - (X_i - x)S_1\}K_h(x - X_i)}{S_0 S_2 - S_1^2}$.

In both cases $\sum_i w_i(x) = 1$. For $p = 1$ also $\sum_i w_i^1(x)(X_i - x) = 0$: the fit reproduces straight lines exactly.

Proof [Supplement]. (i) Minimize $\sum (Y_i - \beta_0)^2 K_h(x - X_i)$.

(ii) $X_x' W_x X_x = \begin{pmatrix} S_0 & S_1 \\ S_1 & S_2 \end{pmatrix}$, and the first row of its inverse is $(S_2, -S_1)/(S_0 S_2 - S_1^2)$. Multiplying by $X_x' W_x$ gives the weights. Summing, $\sum_i (S_2 - (X_i - x)S_1)K_h = S_2 S_0 - S_1^2$, and $\sum_i (S_2 - (X_i - x)S_1)(X_i - x)K_h = S_2 S_1 - S_1 S_2 = 0$. ∎

As $h \to \infty$ the Nadaraya–Watson estimator tends to the constant $\bar{Y}$, while the local linear estimator tends to the least squares line.

Theorem 12.2

Let m be twice continuously differentiable, let the design density f be smooth with $f(x) > 0$, and use the Gaussian kernel. As $n \to \infty$, $h \to 0$ and $nh \to \infty$, conditionally on the X's and

for $p = 0, 1$,

$$\operatorname{bias} \hat{m}_p(x) = B_p(x)h^2 + o_P(h^2), \qquad \operatorname{Var} \hat{m}_p(x) = \frac{\sigma^2(x)}{2\sqrt{\pi} f(x)} \cdot \frac{1}{nh} + o_P\Big(\frac{1}{nh}\Big),$$

with $B_0 = \dfrac{m''}{2} + \dfrac{m'f'}{f}$ and $B_1 = \dfrac{m''}{2}$.

Sketch. Since the weights sum to 1, bias $= \sum_i w_i(x)\{m(X_i) - m(x)\}$ and Var $= \sum_i w_i^2 \sigma^2(X_i)$. For $p = 0$, the numerator $\frac{1}{n}\sum K_h(x - X_i)\{m(X_i) - m(x)\}$ has mean $\int K(u)\{m(x - hu) - m(x)\} f(x - hu)\, du = h^2(\frac{m''f}{2} + m'f') + o(h^2)$. The denominator tends to $f(x)$, giving B_0. For the variance, $\frac{1}{n^2}\sum K_h^2 \sigma^2 \approx \frac{\sigma^2 f}{nh} R(K)$ with $R(\phi) = \frac{1}{2\sqrt{\pi}}$; divide by f^2. For $p = 1$, the property $\sum w_i^1 (X_i - x) = 0$ removes the first-order term $m'(x)(X_i - x)$ exactly, including the asymmetry of the design that produced $m'f'/f$. The leading variance is unchanged. ■

The local linear estimator has the same leading variance as Nadaraya–Watson but no **design bias** $m'f'/f$. It also behaves better near the boundary, which is why it is the default. Generally the bias decreases with the degree, $|\operatorname{bias} \hat{m}_0| \gtrsim |\operatorname{bias} \hat{m}_1| \gtrsim |\operatorname{bias} \hat{m}_2| \gtrsim \dots$. The leading variance follows the pattern

$$\operatorname{Var} \hat{m}_0 \approx \operatorname{Var} \hat{m}_1 < \operatorname{Var} \hat{m}_2 \approx \operatorname{Var} \hat{m}_3 < \cdots .$$

Raising an even degree to the next odd one therefore reduces bias at no cost in variance, which is why odd degrees (usually $p = 1$, sometimes $p = 3$) are preferred. The bias $\frac{m''}{2}h^2$ is positive in valleys and negative at peaks, so the estimate flattens features of the curve.

12.2 Choosing the bandwidth

Motivation. The bandwidth should minimize prediction error, which we cannot observe. Leave-one-out cross-validation estimates it, and for linear smoothers it can be computed from a single fit.

Integrating the bias and variance against the design density,

$$\text{MISE} \approx \frac{h^4}{4} \int (m'')^2 f + \frac{\int \sigma^2}{2\sqrt{\pi}\, nh}, \qquad h^* = \Big(\frac{\int \sigma^2}{2\sqrt{\pi} \int (m'')^2 f \cdot n}\Big)^{1/5}.$$

- **Direct plug-in** (Ruppert–Sheather–Wand): estimate $\int (m'')^2 f$ and $\int \sigma^2$ by blockwise quartic fits, with the number of blocks chosen by an information criterion.
- **Leave-one-out cross-validation:** minimize $\mathrm{CV}(h) = \frac{1}{n}\sum_i \{Y_i - \hat{m}^{(-i)}(X_i)\}^2$.

Proposition 12.3

For a linear smoother whose leave-one-out fit equals the full fit at X_i after deleting point i (true for local polynomial fits), $\mathrm{CV}(h) = \dfrac{1}{n}\sum_i \Big(\dfrac{Y_i - \hat{m}(X_i)}{1 - w_i(X_i)}\Big)^2$.

Proof [Supplement]. Let $\tilde{Y}_i = \hat{m}^{(-i)}(X_i)$. Replacing Y_i by $\tilde{Y}_i$ leaves the weighted fit at X_i unchanged: the added point lies exactly on the leave-one-out local fit, so it does not change the weighted least squares solution. Hence $\tilde{Y}_i = \sum_{j \neq i} w_j(X_i) Y_j + w_i(X_i)\tilde{Y}_i$, so $\tilde{Y}_i = \frac{\hat{m}(X_i) - w_i(X_i) Y_i}{1 - w_i(X_i)}$ and $Y_i - \tilde{Y}_i = \frac{Y_i - \hat{m}(X_i)}{1 - w_i(X_i)}$. ■

So CV needs only one fit per h.

12.3 Confidence intervals

As for densities, intervals target the smoothed curve $m_h(x) = E\hat{m}_1(x)$.

- **Theory-based:** $\widehat{\text{se}}(\hat{m}_1(x)) = \left(\sum_i w_i^1(x)^2 \hat{\sigma}^2(X_i)\right)^{1/2}$ and $m_h(x) \in \hat{m}_1(x) \pm z_{\alpha/2}\widehat{\text{se}}$. Under homoscedasticity, with W the $n \times n$ smoother matrix ($\hat{\mathbf{y}} = W\mathbf{y}$),

$$\hat{\sigma}^2 = \frac{\|(I - W)\mathbf{y}\|^2}{\text{tr}\{(I - W)'(I - W)\}}.$$

  Under heteroscedasticity, smooth $\log r_i^2$ against X_i and exponentiate, which keeps $\hat{\sigma}^2(x) > 0$.
- **Bootstrap:** resample pairs (X_i, Y_i), refit with the *same* h on a grid, and use $\hat{m}_1(x) \pm z_{\alpha/2}\widehat{\text{se}}^*(x)$ or the percentile interval $(Q^*_{\alpha/2}(x), Q^*_{1-\alpha/2}(x))$. Residual bootstraps need care because the residuals inherit the smoothing bias.

Simultaneous bands are possible, as in Section 11.4, but are rarely used in practice.